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The evolution of a black hole's force-free magnetosphere

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SUMMARY

This paper elucidates the structure of a stationary axisymmetric force-free magnetosphere of a Kerr black hole, and the hole's evolution due to extraction of rotational energy by the Blandford–Znajek process. It is argued that there is an 'effective' ergoregion inside the static-limit surface where the densities of 'energy at infinity' and angular momentum of the field are negative. In the outer half of the charged magnetosphere, a pulsar-type centrifugal slingshot wind blows outwards, whereas in the inner half, a similar centrifugal slingshot wind blows *inwards*. The critical conditions for both winds at the fast magnetosonic points near the 'stretched' horizon and the infinity horizon, yield the two expressions for the angular momentum flux per unit flux tube $I(\Psi)$ as a function of stream function Ψ in terms of $(\Omega_H - \Omega_F)$ and Ω_F and respective magnetic fluxes, where Ω_H is the hole's angular velocity and Ω_F is the angular velocity of field line Ψ. Matching the two $I(\Psi)$s at both ends of each field line yields $\Omega_F = \Omega_H/(1+\zeta)$ and $I(\Psi) = [\zeta/2(1+\zeta)]\Omega_H(B_P\varpi^2)_H$, where $\zeta(\Psi) \equiv (B_P\varpi^2)_L/(B_P\varpi^2)_H$ is a parameter expressing degree of collimation of field lines and suffix L denotes the boundary to the 'load region'.

For $\zeta \to 0$, the extraction process is almost adiabatic, and the hole with angular momentum J_H loses its rotational energy, following the path of constant entropy with infinite long time in the J_H–Ω_H plane. For $\zeta \to \infty$, all of the rotational energy is wasted by the surface Joule dissipation and the hole with mass M_H moves on a curve of constant M_H in the J_H–Ω_H plane. The effects of variable ζ upon a hole's magnetospheric structure, wind system and evolution are discussed in some detail.

1 INTRODUCTION

The possibility of extracting a hole's energy was realized by Penrose (1969), who showed that the existence of negative energy orbits within the ergosphere surrounding a Kerr black hole permits the mechanical extraction of energy. Misner, Thorne & Wheeler (1973) gave an example of putting the Penrose process to practical use, to resolve by one stroke the garbage disposal problem and the energy problem in a highly civilized society around a Kerr hole. Christodoulou (1970) then showed that a portion of the rest mass of a Kerr hole can be removed and extracted to infinity, and hence may be regarded as 'reducible'. Christodoulou & Ruffini (1971) next gave the mass formula of a Kerr hole with rest mass M_H, which implies that up to 29 per cent of the total mass energy $M_H c^2$ can be stored as rotational energy. The remainder, larger than 71 per cent, is the 'irreducible' mass, which is thermodynamically interpreted as the hole's entropy (Bekenstein 1975; Hawking 1976). Blandford & Znajek (1977) first studied a full-scale magnetosphere around a rotating black hole. Their aim was to use a stationary axisymmetric force-free magnetosphere to extract sufficient energy from rotating supermassive Kerr holes to power quasars or active galactic nuclei. It was then shown by Phinney (1983) that the original, non-magnetic Penrose process is not adequate in real astrophysical situations, but recently the magnetic Penrose process inside the ergoregion has been studied (Wagh & Dadhich 1989; Punsley & Coroniti 1990).

Meanwhile Thorne & Macdonald (1982) developed the formulation of general relativity based upon the absolute-space/universal-time formalism. Using their '3 + 1' formalism of black hole electrodynamics, Macdonald & Thorne (1982) gave basic equations governing the space outside a rotating axisymmetric black hole, i.e. the black hole magnetosphere and accretion disc, and reformulated and extended the Blandford & Znajek (1977) process. Macdonald (1984) has numerically solved the 'stream equation' for a stationary axisymmetric force-free magnetosphere. Phinney (1982, 1983), on the other hand, investigated the black hole driven hydromagnetic winds. Another significant contribution to black hole astrophysics has been made by Thorne,

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Price & Macdonald (1986), who proposed the membrane paradigm for black holes. Camenzind (1989; references therein) discussed hydrodynamic flows from rapidly rotating compact objects and the formation of relativistic jets.

On the other hand, the analyses of Hawking (1976), Davies (1977) and Tranah & Landsberg (1980) on thermodynamical properties of Kerr black holes were extended by Okamoto & Kaburaki (1990, 1991; hereafter OK-I and OK-II, respectively). It is shown there that thermodynamical state diagrams as they are can be used as the hole's evolutional diagrams. The evolutional paths due to the three extraction processes with $dS_H=0$, $dM_H=0$ and $T_H dS_H=-c^2 dM_H=-\frac{1}{2}\Omega_H dJ_H$ were depicted in OK-I. The last process was shown to be given by the Blandford–Znajek process with $\Omega_F=\frac{1}{2}\Omega_H$, i.e. $\zeta=1$. It will be shown here that the first two cases correspond to $\zeta\to 0$ and $\zeta\to 1$ in the Blandford–Znajek process. The regulation on the accretion process was proposed, to keep the third law of black hole thermodynamics in OK-II. Also, Kaburaki & Okamoto (1991) showed that one can construct a cosmic power station, making use of a black hole Carnot cycle.

The purpose of this paper is to reveal the underlying physical properties in a stationary axisymmetric force-free magnetosphere as well as to clarify the effects of the field structure through parameter ζ upon the hole's evolution. The existence of the accretion disc surrounding the hole is essential in the extraction process of the hole's rotational energy through its magnetosphere, in the sense that it is the dynamo process in the accretion disc that eventually provides magnetic flux to the hole (see Thorne *et al.* 1986). Also, the disc itself may furnish its rotational energy and angular momentum through its own system of magnetosphere and wind to the load region (*cf.* Camenzind 1989). However, our main concern is in the magnetosphere and evolution of a Kerr black hole itself, and hence we do not include the accretion disc explicitly in the present analysis.

We make full use of the '3 + 1' formulation of the black hole electrodynamics and the membrane paradigm. This concept greatly simplifies the present analyses. The general-relativistic effects in Macdonald & Thorne's (1982) '3 + 1' formalism of black hole electrodynamics are condensed into only two scalar functions; ω is the angular velocity of the zero-angular-momentum-observers, or ZAMOs, living in the neighbourhood of a hole and α is their gravitational redshift factor or lapse function. One may thus well refer to the effects due to strong dragging of the inertial frame as the '$\alpha\omega$-mechanism'. It is really the effects of $\alpha\omega$ that give rise to the ingoing centrifugal slingshot wind which corresponds to the 'garbage' thrown down on to the horizon with negative kinetic energy somewhere within the ergosphere, while the outgoing wind corresponds to the returning 'shuttles' (Misner *et al.* 1973). The outer half of the black hole magnetosphere may be attributed to the almost flat space-time, while the inner half is created by the $\alpha\omega$ mechanism. Both halves of the magnetosphere must be antisymmetric to each other in the charge distribution, to preserve charge neutrality of a hole. These are essential properties of the black hole magnetospheres (see Fig. 1).

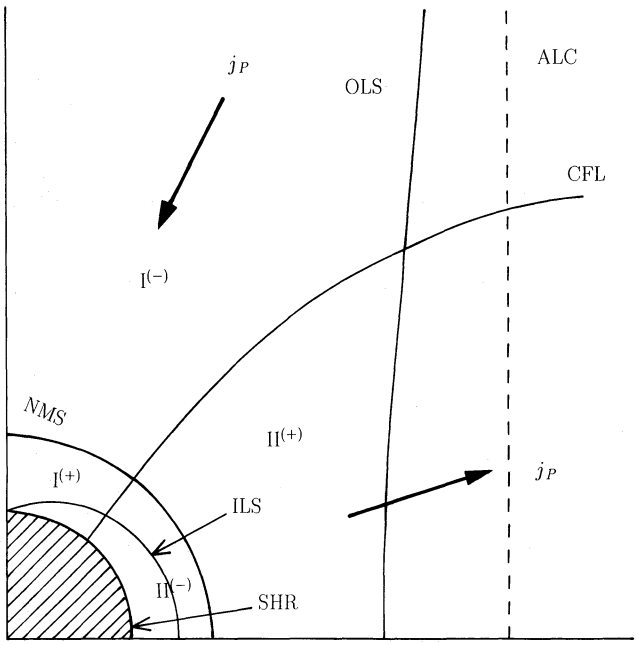


Figure 1. A schematic picture of a force-free magnetosphere in which the angular velocity of field lines equals half of the horizon's angular velocity ($\Omega_F=(1/2)\Omega_H$, i.e. $\zeta=1$). Although the accretion disc is not shown here for simplicity, the analyses in the text are flexible enough to fit any reasonable models of the disc. The meanings of abbreviations are as follows: SHR: the stretched horizon, ILS: the inner light surface, NMS: the null mid-surface, OLS: the outer light surface, ALS: the asymptotic light cylinder, and CFL: the critical field line separating the charge distribution from positive to negative. The electric current flows from infinity to the SHR in the Region-I ($j_P<0$), and flows out of the SHR to infinity in the Region-II ($j_P>0$). The outgoing wind blows from near the NMS and it consists of electrons in the Domain-$I^{(-)}$ ($\varrho_e<0$) and of positrons in the Domain-$II^{(+)}$ ($\varrho_e>0$). The ingoing wind blowing towards the SHR consists of positrons in the Domain-$I^{(+)}$ ($\varrho_e>0$) and of electrons in the Domain-$II^{(-)}$ ($\varrho_e<0$).

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We present basic equations necessary for studying a stationary axisymmetric degenerate force-free magnetosphere of a Kerr hole in Section 2. We derive the expressions for the distributions of the densities of electric charge and current in Section 3, and discuss the asymptotic behaviours of electrodynamical quantities near the infinity horizon in Section 4 and also near the stretched horizon in Section 5. Making reference to pulsar wind theories, we consider the motion of massless plasma in the force-free magnetosphere and derive the critical conditions at the magnetosonic points at the infinity horizon and the stretched horizon in Section 6. We introduce there a parameter function of the stream function, ζ, which not only relates the angular frequency of field lines to that of a hole, but also reflects field topology. Then, taking into account the effects of ζ, we work out the structure of the force-free magnetosphere in Section 7. We present the DC circuit theory of magnetospheric current systems in Section 8. We clarify electromagnetic processes on the two horizons existing for the ZAMOs in Section 9. Evolution of a Kerr hole due to extraction of rotational energy is discussed in Section 10. The final section gives discussion and conclusions.

2 BASIC EQUATIONS

We summarize equations necessary in the following, to treat stationary axisymmetric degenerate force-free magnetospheres. The absolute space around a Kerr hole with mass $M_{\rm H}$ and angular momentum per unit mass $a \equiv J_{\rm H}/M_{\rm H}c$ is described in a Boyer–Lindquist coordinate system;

$$ds^2 = (\rho^2/\Delta)\,dr^2 + \rho^2\,d\theta^2 + \varpi^2\,d\phi^2, \qquad \rho^2 \equiv r^2 + a^2\cos^2\theta, \qquad \Delta \equiv r^2 - 2GM_{\rm H}r/c^2 + a^2, \qquad \Sigma^2 \equiv (r^2+a^2)^2 - a^2\Delta\sin^2\theta,$$
$$\omega = 2aGM_{\rm H}r/c\Sigma^2, \qquad \alpha = \rho\Delta^{1/2}/\Sigma, \qquad \varpi = (\Sigma/\rho)\sin\theta, \qquad \boldsymbol{g} = -(c^2/2\alpha^2)\nabla\alpha^2. \tag{2.1}$$

Here ω is the angular velocity of ZAMOs, α is the lapse function, which is the ratio of the lapse of the ZAMO's proper time $d\tau$ to the lapse of global time dt, and ϖ is the length of a Killing vector $\boldsymbol{m} = \varpi^2\nabla\varphi$. Also ∇ is the covariant derivative operator in absolute space, and $\boldsymbol{g}$ is the gravitational acceleration measured by a ZAMO.

The horizon of the Kerr hole is regarded as a two-dimensional membrane $\alpha = \Delta = 0$ situated at

$$r = r_{\rm H} \equiv GM_{\rm H}/c^2 + [(GM_{\rm H}/c^2)^2 - a^2]^{1/2}. \tag{2.2}$$

Everywhere on the horizon the ZAMOs are dragged into the uniform rotation of the hole, i.e.

$$\omega \to \Omega_{\rm H} \equiv c^3 a/2GM_{\rm H}r_{\rm H}. \tag{2.3}$$

They also feel an infinitely large gravitational acceleration, $|\boldsymbol{g}| \to \infty$ except for in an extreme Kerr hole, but converted from ZAMOs' proper time to global time, one has

$$\alpha\boldsymbol{g} \to -g_{\rm H}\boldsymbol{n}, \qquad g_{\rm H} \equiv c^4(r_{\rm H} - GM_{\rm H}/c^2)/2GM_{\rm H}r_{\rm H}, \tag{2.4}$$

where $\boldsymbol{n}$ is a unit normal vector pointing orthogonally out of the horizon and $g_{\rm H}$ is the surface gravity of the Kerr hole. For an extreme Kerr hole with $r_{\rm H} = a = GM_{\rm H}/c^2$ in (2.2), one sees that the surface gravity vanishes and therefore the redshifted temperature tends to zero. In this case, $\boldsymbol{g}$ no longer tends to infinity toward the horizon, but remains finite, possessing the θ-component. One thus needs some cautious treatment for an extreme Kerr hole, related to the problem of vanishing magnetic flux. Introducing a coordinate system (α, λ, ϕ) in calculations near the horizon, the metric of absolute space reads

$$ds^2 = (c^2/g_{\rm H})^2\,d\alpha^2 + d\lambda^2 + \varpi^2\,d\phi^2, \tag{2.5}$$

where $d\lambda = \rho\,d\theta$. Then the element of the proper area and the total area on the horizon surface are given by

$$dA_{\rm H} = \varpi_{\rm H}\,d\lambda\,d\phi = (r_{\rm H}^2 + a^2)\sin\theta\,d\theta\,d\phi, \qquad A_{\rm H} = 4\pi(r_{\rm H}^2 + a^2). \tag{2.6a,b}$$

Here all the electromagnetic quantities such as the electric field $\boldsymbol{E}$, the magnetic field $\boldsymbol{B}$, the electric charge density $\varrho_{\rm e}$ and the current density $\boldsymbol{j}$ are measured by the ZAMOs. In stationary axisymmetric electrodynamics these are given in terms of the three independent, freely specifiable scalar functions A_0, Ψ and I, where A_0 is the scalar potential, Ψ measures the total magnetic flux passing upward through an $\boldsymbol{m}$-loop and is the so-called stream function, related to the component A_ϕ of the vector potential $\boldsymbol{A}$ as $\Psi = 2\pi A_\phi = 2\pi\boldsymbol{A}\cdot\boldsymbol{m}$, and I is the total current passing downward through an $\boldsymbol{m}$-loop.

The assumption of degeneracy, which is applicable to regions of space where $|\boldsymbol{B}^2 - \boldsymbol{E}^2| \gg |\boldsymbol{E}\cdot\boldsymbol{B}| \simeq 0$, allows the use of two specifiable scalar functions I, Ψ and one function of Ψ, $A_0 = A_0(\Psi)$. The derivative of A_0 with respect to Ψ, defined as

$$\Omega_{\rm F}(\Psi) = -2\pi c(dA_0/d\Psi) \tag{2.7}$$

has the physical meaning of the angular velocity of the magnetic field lines relative to absolute space. The toroidal velocity relative to the ZAMOs is then given by

$$\boldsymbol{v}_{\rm F} = [(\Omega_{\rm F} - \omega)/\alpha]\,\boldsymbol{m}. \tag{2.8}$$

As in the case of the pulsar magnetospheres, let us consider a region of the magnetosphere, where sufficient amounts of plasma with perfect conductivity are available and one can approximate the fields there as not only degenerate but force-free, i.e.

$$\varrho_{\rm e}\boldsymbol{E} + (\boldsymbol{j}/c)\times\boldsymbol{B} \simeq 0. \tag{2.9}$$

Then, in the force-free regions the toroidal component of (2.9) yields $\nabla\Psi\times\nabla I = 0$, and hence $I = I(\Psi)$. One can thus express the

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electromagnetic quantities in terms of Ψ, $\Omega_F(\Psi)$ and $I(\Psi)$ as follows (*cf.* Macdonald & Thorne 1982):

$$\boldsymbol{B}_P = -(1/2\pi\varpi^2)(\boldsymbol{m}\times\nabla\Psi), \qquad \boldsymbol{B}_T = -(2I/\alpha c\varpi^2)\boldsymbol{m}, \tag{2.10a,b}$$

$$\boldsymbol{E}_P = -(\boldsymbol{v}_F/c)\times\boldsymbol{B}_P = -[(\Omega_F-\omega)/2\pi\alpha c]\nabla\Psi, \tag{2.10c}$$

$$\boldsymbol{j}_P = -(1/\alpha)(dI/d\Psi)\boldsymbol{B}_P, \tag{2.10d}$$

$$8\pi^2 j_T = -(c\varpi/\alpha)\nabla\cdot[(\alpha/\varpi^2)\nabla\Psi] - [\varpi(\Omega_F-\omega)/\alpha^2 c]\nabla\omega\cdot\nabla\Psi, \tag{2.10e}$$

$$8\pi^2\varrho_e = -\nabla\cdot\{[\varpi^2(\Omega_F-\omega)/\alpha^2 c][(\alpha/\varpi^2)\nabla\Psi]\}, \tag{2.10f}$$

where $\boldsymbol{E}_T \equiv 0$. One can interpret $-\varpi\alpha B_T/4\pi = I(\Psi)/2\pi c$ as the rate per unit global time of transport of angular momentum along a unit flux tube [*cf.* Mestel 1968; see also (2.18b) later]. The poloidal component of (2.9) yields

$$j_T = [\varpi(\Omega_F-\omega)/\alpha]\varrho_e + (1/\alpha^2 c\varpi)(dI^2/d\Psi), \tag{2.11}$$

which combines with (2.10d) to yield

$$\boldsymbol{j} = \varrho_e\boldsymbol{v}_F - (1/\alpha)(dI/d\Psi)\boldsymbol{B}. \tag{2.12}$$

Substituting (2.10e,f) into (2.11) yields a so-called stream equation

$$\nabla\cdot\left\{\frac{\alpha}{\omega^2}\left[1-\left(\frac{\varpi(\Omega_F-\omega)}{\alpha c}\right)^2\right]\nabla\Psi\right\} + \frac{\Omega_F-\omega}{\alpha c^2}\frac{d\Omega_F}{d\Psi}|\nabla\Psi|^2 + \frac{8\pi^2}{\alpha\varpi^2 c^2}\frac{dI^2}{d\Psi} = 0 \tag{2.13}$$

(Blandford & Znajek 1977; Macdonald & Thorne 1982; Bekenstein & Eichler 1985). Using an iterative relaxation method on a region bounded by the hole's horizon, the accretion disc and an outer boundary beyond which the force-free condition breaks down, Macdonald (1984) solved (2.13) with a Neuman-type boundary condition

$$\nabla\Psi\cdot\nabla\left\{\frac{\alpha}{\varpi^2}\left[1-\left(\frac{\varpi(\Omega_F-\omega)}{\alpha c}\right)^2\right]\right\} + \frac{\Omega_F-\omega}{\alpha c^2}\frac{d\Omega_F}{d\Psi}|\nabla\Psi|^2 + \frac{8\pi^2}{\alpha\varpi^2 c^2}\frac{dI^2}{d\Psi} = 0 \tag{2.14a}$$

at the two light surfaces, i.e. at the outer and inner light surfaces

$$\varpi_{OL} = \alpha c/(\Omega_F-\omega), \qquad \varpi_{IL} = \alpha c/(\omega-\Omega_F). \tag{2.14b}$$

The present purpose is to clarify basic physical properties that a plausible solution for the force-free regions of a magnetosphere must possess.

If one defines

$$\tilde{\boldsymbol{B}} = (1-v_F^2/c^2)\boldsymbol{B}_P + \boldsymbol{B}_T, \tag{2.15a}$$

similarly to the pulsar case (Mestel 1973), after some manipulation one has

$$(c/4\pi)\nabla\times\alpha\tilde{\boldsymbol{B}} = -(dI/d\Psi)\boldsymbol{B} + (\varpi^2 B_P^2/\alpha c)(d\Omega_F/d\Psi)\boldsymbol{v}_F, \tag{2.15b}$$

in the vectorial form. The toroidal component of (2.15b) reproduces the stream equation (2.13) and the poloidal component is just an identity relation. From (2.12) and (2.15b) for the electric current one obtains

$$\boldsymbol{j} = (c/4\pi\alpha)\nabla\times\alpha\tilde{\boldsymbol{B}} + [\varrho_e - (\varpi^2 B_P^2/2\alpha c)(d\Omega_F/d\Psi)]\boldsymbol{v}_F. \tag{2.15c}$$

If the field lines emanating from the stretched horizon and reaching the acceleration zone are constrained to rotate uniformly, one has

$$(c/4\pi)\nabla\times\alpha\tilde{\boldsymbol{B}} = -(dI/d\Psi)\boldsymbol{B}, \qquad \boldsymbol{j} = (c/4\pi\alpha)\nabla\times\alpha\tilde{\boldsymbol{B}} + \varrho_e\boldsymbol{v}_F. \tag{2.16a,b}$$

For later convenience, using Macdonald & Thorne's (1982) formulae (2.30a,b) and (2.31a,b) for densities and fluxes of redshifted energy and angular momentum for electromagnetic field, one can compute for the stationary axisymmetric degenerate force-free case;

$$\varepsilon_E = (\alpha/8\pi)B_P^2[1+\varpi^2(\Omega_F^2-\omega^2)/\alpha^2 c^2 + (2I/\alpha c\varpi B_P)^2], \tag{2.17a}$$

$$\varepsilon_J = (1/c)[(\Omega_F-\omega)B_P\varpi^2/4\pi\alpha c]B_P, \tag{2.17b}$$

$$\boldsymbol{S}_E^P = (\alpha c/4\pi)(\boldsymbol{E}\times\boldsymbol{B})_P + \omega\boldsymbol{S}_J^P = (I/2\pi)[(\omega/\alpha c)\boldsymbol{B}_P - (1/\varpi^2)(\boldsymbol{E}\times\boldsymbol{m})] = \Omega_F(I/2\pi\alpha c)\boldsymbol{B}_P, \tag{2.18a}$$

$$\boldsymbol{S}_J^P = (I/2\pi\alpha c)\boldsymbol{B}_P, \tag{2.18b}$$

where (2.10a,b,c) are substituted. The toroidal components of $\boldsymbol{S}_E$ and $\boldsymbol{S}_J$ are dropped for brevity in (2.18a,b). It can be seen in (2.18a,b) that $\Omega_F I/2\pi c$ and $I/2\pi c$ denote energy and angular momentum fluxes per unit magnetic flux tube per unit global time along the poloidal field lines, and are functions of Ψ only. This implies that if one takes $I(\Psi)>0$ for a black hole, the poloidal components of $\boldsymbol{S}_E$ and $\boldsymbol{S}_J$ point outward all the way from the horizon to infinity, although the Poynting flux $(\boldsymbol{E}\times\boldsymbol{B})_P$ points inwardly inside a surface of $\omega=\Omega_F$ [*cf.* (3.7)]. On the other hand, the densities of energy and angular momentum become

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negative near the horizon (see Section 7.4). It is easy to note $\nabla\cdot(\alpha \boldsymbol{S}_E)=\nabla\cdot(\alpha \boldsymbol{S}_J)=0$. The conditions specifying $\Omega_F(\Psi)$ and $I(\Psi)$ along a given field line shall be described later on the basis of current stellar wind theory (see Section 6).

3 THE DISTRIBUTIONS OF ELECTRIC CHARGE DENSITY AND ELECTRIC CURRENT

The electric charge density ϱ_e and the toroidal component j_T of the electric current vector expressed in (2.10e,f) contain the first and second derivatives of Ψ. It is then convenient to eliminate $\nabla\cdot[(\alpha/\varpi^2)\nabla\Psi]$ between (2.10e,f), to relate ϱ_e to j_T as follows:

$$\varrho_e=\frac{\varpi(\Omega_F-\omega)}{\alpha c}\left[j_T-\frac{c^2}{16\pi^2\varpi}\nabla\Psi\cdot\boldsymbol{G}\right], \tag{3.1a}$$

$$\boldsymbol{G}\equiv\frac{c^2}{2}\left[\nabla\ln\frac{\varpi^2}{\alpha^2}(\Omega_F-\omega)-\frac{\varpi^2(\Omega_F-\omega)}{\alpha^2c^2}\nabla\omega\right]=\boldsymbol{g}+c^2\nabla\ln\varpi+\frac{c^2}{2}\frac{(d\Omega_F/d\Psi)}{\Omega_F-\omega}\nabla\Psi-\frac{c^2}{2}\left[1+\frac{\varpi^2(\Omega_F-\omega)^2}{\alpha^2c^2}\right]\frac{\nabla\omega}{\Omega_F-\omega}. \tag{3.1b}$$

Combining (3.1a) with (2.11), one can solve for ϱ_e and j_T;

$$\varrho_e=\frac{\Omega_F-\omega}{4\pi^2\alpha c^3}\frac{(4\pi^2/\alpha^2)(dI^2/d\Psi)-\nabla\Psi\cdot\boldsymbol{G}}{1-[\varpi(\Omega_F-\omega)/\alpha c]^2}, \tag{3.2a}$$

$$j_T=\frac{1}{4\pi^2c\varpi}\frac{(4\pi^2/\alpha^2)(dI^2/d\Psi)-[\varpi(\Omega_F-\omega)/\alpha c]^2(\nabla\Psi\cdot\boldsymbol{G})}{1-[\varpi(\Omega_F-\omega)/\alpha c]^2}. \tag{3.2b}$$

In the flat space limit of $\alpha\to1$ and $\omega\to0$ in (3.2a,b), one has the corresponding pulsar force-free equations [see (44), (45) and (46) in Okamoto (1974)]. The significance of (3.2a,b) as compared with (2.10e,f) lies in the fact that even if stream equation (2.13) is not explicitly solved, one can work out the effects of the two functions $I(\Psi)$ and Ω_F upon the behaviours of ϱ_e and j_T along each field line.

Let us at first presume that the total electric current flowing into and out of the stretched horizon vanishes in the stationary state, and then from the definition of $I(\Psi)$ one has, from (2.10d) and (2.6a),

$$\oint\alpha\boldsymbol{j}_P\cdot\boldsymbol{n}\,dA_H=\oint\left(-\frac{dI}{d\Psi}\right)\cdot2\pi\varpi_H B_n\,d\lambda=-[I(\Psi_{eq})-I(\Psi_0)]=0, \tag{3.3}$$

where B_n is the normal component of the magnetic field given by $d\Psi=2\pi\varpi_H B_n\,d\lambda$. The integration is taken over the stretched horizon surface from the north pole where $\Psi=\Psi_0$ to a certain limiting field line open to the outer world, or here for simplicity to the equator, assuming antisymmetry between both the north and south hemispheres. Condition (3.3) indicates the presence of some critical field line Ψ_c such that

$$dI/d\Psi>0 \text{ in } \Psi_0<\Psi<\Psi_c;\qquad dI/d\Psi=0 \text{ at } \Psi=\Psi_c;\qquad dI/d\Psi<0 \text{ in } \Psi_c<\Psi<\Psi_{eq}, \tag{3.4}$$

and that the angular momentum transport per unit flux tube $I(\Psi)$ reaches its maximum value at $\Psi=\Psi_c$. It can then be seen in (3.4) and (2.10d) that the electric current flows inwardly along the field lines with $\Psi_0<\Psi<\Psi_c$ from the acceleration region to the stretched horizon and outwardly along those with $\Psi_c<\Psi<\Psi_{eq}$ from the stretched horizon to the acceleration region (see Fig. 2). Along the critical field line $\Psi=\Psi_c$, one finds in (2.10d), (2.11) and (3.2a);

$$\boldsymbol{j}_P=0,\qquad j_T=\frac{\varpi(\Omega_F-\omega)}{\alpha}\varrho_e,\qquad \varrho_e=-\frac{\Omega_F-\omega}{4\pi^2\alpha c^2}\frac{\nabla\Psi\cdot\boldsymbol{G}}{1-[\varpi(\Omega_F-\omega)/\alpha c]^2}. \tag{3.5a,b,c}$$

The critical field line in the force-free treatment will also have to be a charge-separating field line, and then one obtains $\varrho_e=\nabla\Psi\cdot\boldsymbol{G}=0$ for $\Psi=\Psi_c$, which implies that the poloidal field $\boldsymbol{B}_P$ must be parallel to the vector $\boldsymbol{G}$. We shall discuss the topology of the critical field line later in Section 7.5.

At the two light surfaces where the denominators of ϱ_e and j_T vanish, the numerators must also vanish, i.e. one has one of the so-called critical conditions

$$\frac{4\pi^2}{\alpha^2}\frac{dI^2}{d\Psi}=\nabla\Psi\cdot\boldsymbol{G}. \tag{3.6}$$

It is easy to see that condition (3.6) is equivalent to the boundary condition (2.14a) to the stream equation. Between the two light surfaces $\varpi=\varpi_{IL}$ and $\varpi=\varpi_{OL}$ there is a surface (say ϖ_N) where

$$\omega=\Omega_F,\qquad \boldsymbol{v}_F=\boldsymbol{E}_P=0,$$

$$\varrho_e=-[\nabla\Psi\cdot\nabla(\Omega_F-\omega)]/8\pi^2\alpha c,\qquad j_T=(1/\alpha^2c\varpi)(dI^2/d\Psi), \tag{3.7}$$

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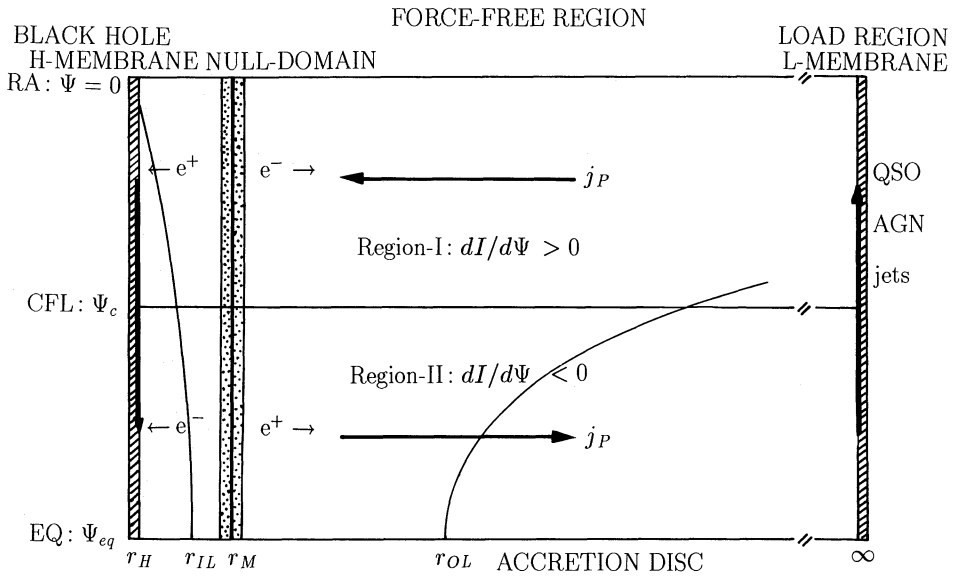


Figure 2. A quarter of a force-free region. The abscissa is the radius r along each field line with Ψ and the ordinate is the stream function Ψ at the true horizon ($r_{\rm H}$), on which there is a thin boundary layer of thickness $\alpha_{\rm H}$.

from (2.8), (2.10c,f) and (2.11). This is the surface that divides the force-free magnetosphere into two, each magnetosphere possessing its own light surface through which the plasma flows inward or outward (Blandford & Znajek 1977). We refer to it as the null surface, because there must be the creation zone of more or less neutral plasma, or the spark gaps, situated in its neighbourhood. The velocity of each field line relative to the ZAMOs emanating from the stretched horizon begins to increase from zero at $\varpi = \varpi_{\rm N}$ and equals c at $\varpi = \varpi_{\rm OL}$, tending to infinity like $\boldsymbol{v}_{\rm F} \simeq \Omega_{\rm F}\boldsymbol{m}$ far from the outer light surface where $\omega \to 0$, and $\alpha \to 1$, while it changes sign at $\varpi = \varpi_{\rm N}$, equals $-c$ and decreases to negative infinity like $\boldsymbol{v}_{\rm F} \simeq -[(\Omega_{\rm H} - \Omega_{\rm F})/\alpha]\boldsymbol{m}$ toward the stretched horizon where $\alpha \to 0$ and $\omega \simeq \Omega_{\rm H}$. Thus the ZAMOs in the outer magnetosphere see the centrifugal magnetic slingshot wind blowing outward from the vicinity of the null surface to the acceleration zone, and the ZAMOs in the inner magnetosphere find the centrifugal magnetic slingshot wind blowing inward to the stretched horizon. The ingoing wind is antisymmetric to the outgoing wind in direction and the associated charge distribution as shown in the next two sections.

4 ASYMPTOTIC BEHAVIOURS OF ELECTRODYNAMIC QUANTITIES NEAR THE LOAD REGION

In this section the asymptotic behaviours of physical quantities are investigated far from the outer light surface, that is, in the boundary area of the pulsar-type wind zone to the load region. Assuming $\alpha = 1$ and $\omega = 0$ there, one has from (2.8) and (2.10a,b,...,d);

$$\boldsymbol{v}_{\rm F} \simeq \Omega_{\rm F}\varpi\boldsymbol{e}_{\phi}, \qquad \boldsymbol{B}_{\rm P} \simeq (\nabla\Psi \times \boldsymbol{e}_{\phi})/2\pi\varpi, \qquad \boldsymbol{B}_{\rm T} \simeq -(2I/c\varpi)\boldsymbol{e}_{\phi}, \qquad \boldsymbol{E} \simeq -(\Omega_{\rm F}/2\pi c)\nabla\Psi, \qquad \boldsymbol{j}_{\rm P} \simeq -(dI/d\Psi)\boldsymbol{B}_{\rm P}, \quad (4.1a,b,c,d,e)$$

where $\varpi = r\sin\theta$ and $\boldsymbol{e}_{\phi}$ is the unit toroidal vector. Also equations (3.2a,b) reduce to

$$\varrho_{\rm e} \simeq \frac{\Omega_{\rm F}}{4\pi^2 c^3}\,\frac{(4\pi^2/c^2)(dI^2/d\Psi) - \nabla\Psi\cdot\boldsymbol{G}}{1-\Omega_{\rm F}^2\varpi^2/c^2},$$

$$j_{\rm T} \simeq \frac{1}{4\pi^2 c\varpi}\,\frac{(4\pi^2/c^2)(dI/d\Psi) - (\Omega_{\rm F}^2\varpi^2/c^2)\nabla\Psi\cdot\boldsymbol{G}}{1-\Omega_{\rm F}^2\varpi^2/c^2}, \qquad (4.2a,b)$$

where $\boldsymbol{G} = (c^2/2)\nabla \ln \Omega_{\rm F}\varpi^2$. For $\varpi \gg \varpi_{\rm OL}$, from (4.2a,b) one obtains

$$\varrho_{\rm e} \simeq -\frac{1}{c\Omega_{\rm F}\varpi^2}\left\{\frac{dI^2}{d\Psi} + \frac{c^2}{\Omega_{\rm F}\varpi^2}\left[\frac{dI^2}{d\Psi} - \frac{\Omega_{\rm F}B_z\varpi^2}{2\pi} - \frac{1}{2}\,\frac{d\ln\Omega_{\rm F}}{d\Psi}(\Omega_{\rm F}B_{\rm P}\varpi^2)^2\right]\right\}, \qquad (4.3a)$$

$$j_{\rm T} \simeq -\frac{c}{\Omega_{\rm F}\varpi^3}\left[\frac{dI^2}{d\Psi} - \frac{\Omega_{\rm F}^2 B_z\varpi^2}{2\pi} - \frac{1}{2}\,\frac{d\ln\Omega_{\rm F}}{d\Psi}(\Omega_{\rm F}B_{\rm P}\varpi^2)^2\right], \qquad (4.3b)$$

because $\nabla\Psi\cdot\boldsymbol{G} = 2\pi c^2 B_z + 2(d\ln\Omega_{\rm F}/d\Psi)(\pi c B_{\rm P}\varpi)^2$ and B_z is the z-component of magnetic field. It can be seen later [*cf.* (4.6) and (4.7b)] that both $B_{\rm P}\varpi^2$ and $B_z\varpi^2$ are of order unity or smaller than $(\varpi_{\rm OL}/\varpi)$. Taking the divergence of $\boldsymbol{E}$ in spherical

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polar coordinates yields

$$\nabla\cdot\boldsymbol{E}=-\frac{1}{2\pi c}\left[\frac{\Omega_{\rm F}}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\Psi}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\Omega_{\rm F}\sin\theta\frac{\partial\Psi}{\partial\theta}\right)+\frac{d\Omega_{\rm F}}{d\Psi}\left(\frac{\partial\Psi}{\partial r}\right)^2\right]. \tag{4.4}$$

Then, using Poisson's equation for charge conservation, one has from (4.3a) and (4.4);

$$\Omega_{\rm F}\sin^2\theta\frac{\partial}{\partial r}\left(r^2\frac{\partial\Psi}{\partial r}\right)+\frac{1}{2\Omega_{\rm F}\partial\Psi/\partial\theta}\frac{\partial}{\partial\theta}\left[\left(\Omega_{\rm F}\sin\theta\frac{\partial\Psi}{\partial\theta}\right)^2-(4\pi I)^2\right]+\frac{d\Omega_{\rm F}}{d\Psi}\left(\varpi\frac{\partial\Psi}{\partial r}\right)^2 \tag{4.5a}$$

$$\simeq\frac{8\pi^2\varpi_{\rm OL}^2}{\Omega_{\rm F}\varpi^2}\left[\frac{dI^2}{d\Psi}-\frac{\Omega_{\rm F}^2B_z\varpi^2}{2\pi}-\frac{1}{2}\frac{d\ln\Omega_{\rm F}}{d\Psi}(\Omega_{\rm F}B_{\rm P}\varpi^2)^2\right]$$

[*cf.* equation (6.7) in Macdonald & Thorne (1982) and (5.8) in this paper]. It can be seen that the right-hand side of (4.5a) is $O(\varpi^{-2})$, while the left-hand side is $O(1)$, and one must therefore require the left-hand side to vanish in the region of $\varpi\gg\omega_{\rm OL}$, that is, one obtains

$$\frac{\partial^2\Psi}{\partial r^2}+\frac{2}{r}\frac{\partial\Psi}{dr}+\frac{d\ln\Omega_{\rm F}}{d\Psi}\left(\frac{\partial\Psi}{\partial r}\right)^2+\frac{1}{2\Omega_{\rm F}^2\varpi^2\partial\Psi/\partial\theta}\frac{\partial}{\partial\theta}\left[\left(\Omega_{\rm F}\sin\theta\frac{\partial\Psi}{\partial\theta}\right)^2-(4\pi I)^2\right]\simeq 0. \tag{4.5b}$$

Let us now expand the stream equation far from the outer light surface into the form

$$\Psi(r,\theta)=\Psi_{\rm L}^{(0)}(\theta)+\Psi_{\rm L}^{(1)}(\theta)\ln r+\Psi_{\rm L}^{(2)}(\theta)/r+O(r^{-2}), \tag{4.6}$$

and examine the following two cases.

In the first case one requires Ψ to be non-singular far from the light surface, i.e. $\Psi_{\rm L}^{(1)}(\theta)\equiv 0$, and then the absence of the $\ln r$ term from Ψ implies that the field line shape becomes asymptotically radial toward the boundary area to the load region. In this case the field line shape does not depend explicitly upon whether the field lines are rotating uniformly or differentially near infinity. Also the absence is equivalent to the demand in (4.5b) that

$$I(\Psi)=\pm(1/4\pi)\Omega_{\rm F}\sin\theta[d\Psi_{\rm L}^{(0)}(\theta)/d\theta]=\pm(1/2)\Omega_{\rm F}(B_{\rm P}\varpi^2)_{\rm L},$$

$$(B_{\rm P}\varpi^2)_{\rm L}=(1/2\pi)\sin\theta[d\Psi_{\rm L}^{(0)}(\theta)/d\theta]. \tag{4.7a,b}$$

It can be seen in Section 6 that condition (4.7a) is the critical condition itself, derived at the magnetosonic point at infinity.

In the second case, if one adopts the $\ln r$ term in (4.6) and takes $\Psi_{\rm L}^{(0)}(\theta)\equiv 0$, requiring the first three terms to vanish in (4.5b), one obtains

$$\Psi_{\rm L}^{(1)}(\theta)=-1/(d\ln\Omega_{\rm F}/d\Psi), \tag{4.8}$$

which combines with (4.1b,d) to yield

$$B_r\simeq(\ln r/r^2)(1/2\pi\sin\theta)[d\Psi_{\rm L}^{(1)}(\theta)/d\theta)], \tag{4.9a}$$

$$B_\theta\simeq-(1/2\pi r^2\sin\theta)\Psi_{\rm L}^{(1)}(\theta)=(\Omega_{\rm F}/2\pi r^2\sin\theta)(d\Psi/d\Omega_{\rm F}), \tag{4.9b}$$

$$E_r\simeq(\Omega_{\rm F}^2/2\pi cr)/(d\Omega_{\rm F}/d\Psi),\qquad E_\theta\simeq-(1/2\pi c)(\ln r/r)[d\Psi_{\rm L}^{(1)}(\theta)/d\theta]. \tag{4.10a,b}$$

If one still requires the fourth term of (4.5b) to vanish for $\Psi=\Psi_{\rm L}^{(1)}(\theta)\ln r$, one obtains

$$4\pi I(\Psi)/\Omega_{\rm F}=\pm\ln r[d\Psi_{\rm L}^{(1)}/d\theta]\sin\theta. \tag{4.11}$$

In the second case the field topology is directly coupled with the differential rotation of field lines. If $\Omega_{\rm F}$ is a decreasing function of Ψ, i.e. $d\Omega_{\rm F}/d\Psi<0$, one can see from (4.9a,b) and (4.10a,b) that the magnetic field line will be bent poleward in the sense of collimation to the rotation axis ($B_\theta<0$) and the radial electric field is inwardly directed ($E_r<0$), B_r and E_θ being dependent upon the sign of $d\Psi_{\rm L}^{(1)}(\theta)/d\theta$. It will need further detailed study to clarify whether the second case may yield reasonable critical conditions and plausible field topology in the neighbourhood of the boundary area to the load region. One will, however, certainly have to take into account not only differential rotation and collimation of field lines but also particle inertia, to reach a satisfactory solution. Meanwhile we will use the non-singular solution without the $\ln r$ term in (4.6), together with the critical condition (4.7a,b).

One should assume the (+) sign for I in the present context of the black hole magnetosphere, and the physical meaning of the choice of (− versus +) sign will be discussed later in connection with the similar condition to (4.7a) near the stretched horizon. Then for the asymptotic forms of the electromagnetic quantities one has from (4.1c,d,e), (4.3a) and (4.7a,b);

$$B_{\rm T}\simeq-\frac{2I}{c\varpi}=-\frac{\Omega_{\rm F}}{c\varpi}(B_{\rm P}\varpi^2)_{\rm L},\qquad E_\theta\simeq-\frac{\Omega_{\rm F}}{2\pi cr}\frac{d\Psi_{\rm L}^{(0)}}{d\theta}=-\frac{\Omega_{\rm F}}{c\varpi}(B_{\rm P}\varpi^2)_{\rm L},\qquad \varrho_{\rm e}\simeq-\frac{1}{c\Omega_{\rm F}\varpi^2}\frac{dI^2}{d\Psi}=-\frac{(B_{\rm P}\varpi^2)_{\rm L}}{c\varpi^2}\frac{dI}{d\Psi}, \tag{4.12a,b,c,d}$$

$$j_r\simeq-\frac{(B_{\rm P}\varpi^2)_{\rm L}}{\varpi^2}\frac{dI}{d\Psi}\simeq c\varrho_{\rm e}$$

[*cf.* (92), (93) and (94) in Okamoto (1974)].

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These results reveal several interesting features in the neighbourhood of infinity. Equations (4.12a,b) emanating from the condition (4.7a) with positive sign indicate that the electromagnetic fields satisfy the relation

$$\boldsymbol{B}_{\mathrm{T}}=\boldsymbol{n}\times\boldsymbol{E}_{\theta},\qquad \boldsymbol{E}_{\theta}=-\boldsymbol{n}\times\boldsymbol{B}_{\mathrm{T}}. \tag{4.13}$$

(*cf.* Mestel *et al.* 1985; Blandford 1989). This is nothing but the 'outgoing-wave' or 'radiative' boundary condition which, similarly to the case in the stretched horizon, allows one to presume the surface electric current on the 'sphere at infinity' (Phinney 1983) or a 'membrane' in the force-free approximation (see Thorne *et al.* 1986). Equations (4.12c,d) show that $\varrho_{\mathrm{e}}<0$, $j_r<0$ in the region with field lines in $\Psi_0<\Psi<\Psi_{\mathrm{c}}$ (Domain-I$^{(-)}$) and $\varrho_{\mathrm{e}}>0$, $j_r>0$ in the region with those in $\Psi_{\mathrm{c}}<\Psi<\Psi_{\mathrm{eq}}$ (Domain-II$^{(+)}$). One can then see that the electric current is almost radial, i.e. $|j_r|\gg|j_{\mathrm{T}}|$, and reaches the speed of light. The electric current is therefore a convection current and the plasma is charge-separated, with electrons outgoing in Domain-I$^{(-)}$ and with positrons (or positive ions if possible) outgoing in Domain-II$^{(+)}$.

5 THE ASYMPTOTIC BEHAVIOURS OF ELECTRODYNAMICAL QUANTITIES NEAR THE STRETCHED HORIZON

This section is devoted to clarifying the situation far inside the inner light surface. The ZAMOs living near the stretched horizon where $\alpha\ll\varpi(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})/c$ are rotating more rapidly than the field lines, and therefore as viewed from the ZAMOs the physical velocity of the field lines is in the reverse direction and its absolute value is much larger than the velocity of light, i.e. from (2.8)

$$\boldsymbol{v}_{\mathrm{F}}\simeq-\alpha^{-1}(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})\boldsymbol{m}\ll-ce_{\phi}. \tag{5.1}$$

Then the magnetic slingshot effect due to the $\alpha\omega$-mechanism is so powerful that the poloidal velocity of massless particles approaches the velocity of light near the horizon.

The poloidal electromagnetic fields near the horizon are in the coordinate system (α,λ,ϕ);

$$\boldsymbol{B}_{\mathrm{P}}=\frac{1}{2\pi\varpi}\left(\frac{\partial\Psi}{\partial\lambda},\ -\frac{g_{\mathrm{H}}}{c^2}\frac{\partial\Psi}{\partial\alpha},\ 0\right),\qquad \boldsymbol{E}_{\mathrm{P}}=\frac{\omega-\Omega_{\mathrm{F}}}{2\pi c\alpha}\left(\frac{g_{\mathrm{H}}}{c^2}\frac{\partial\Psi}{\partial\alpha},\ \frac{\partial\Psi}{\partial\lambda},\ 0\right). \tag{5.2a,b}$$

The expressions for $\nabla\cdot\boldsymbol{E}$, ϱ_{e} and j_{T} in the neighbourhood of the horizon are given in Appendix A. It can be seen in (2.10d) and (A3a) that, in the order of α^{-1}, relations

$$\varrho_{\mathrm{e}}\simeq\alpha^{-1}\frac{dI^2/d\Psi}{c\varpi^2(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})},\qquad \boldsymbol{j}_{\mathrm{P}}\simeq-c\varrho_{\mathrm{e}}\frac{\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}}}{2I}(\boldsymbol{B}_{\mathrm{P}}\varpi^2) \tag{5.3a,b}$$

must hold, and hence very near the horizon one sees that $\varrho_{\mathrm{e}}>0$, $\boldsymbol{j}_{\mathrm{P}}<0$ in the region with field lines in $\Psi_0<\Psi<\Psi_{\mathrm{c}}$ (Domain-I$^{(+)}$), where $dI/d\Psi>0$, and that $\varrho_{\mathrm{e}}<0$, $\boldsymbol{j}_{\mathrm{P}}>0$ in the region with those in $\Psi_{\mathrm{c}}<\Psi<\Psi_{\mathrm{eq}}$ (Domain-II$^{(-)}$), where $dI/d\Psi<0$. For the choice of the same $(+)$ sign for I as in the case of (4.7a), the direction of the poloidal electric current is the same along a given field line from the infinity horizon to the hole's horizon, whereas the sign of charge density is opposite between both ends of the field line (see Fig. 2).

The equation of charge conservation near the horizon yields, from (A2a,b) and (A3a,b);

$$(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})\alpha\frac{\partial}{\partial\alpha}\left(\frac{1}{\alpha}\frac{\partial\Psi}{\partial\alpha}\right)+\frac{c^4}{g_{\mathrm{H}}^2(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})}\frac{1}{2\varpi^2(\partial\Psi/\partial\lambda)}\frac{\partial}{\partial\lambda}\{[\varpi(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})(\partial\Psi/\partial\lambda)]^2-(4\pi I)^2\}\simeq\frac{d\Omega_{\mathrm{F}}}{d\Psi}\left(\frac{\partial\Psi}{\partial\alpha}\right)^2 \tag{5.4}$$

$$-\frac{c^2}{(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})\varpi^2}\left(\alpha\frac{\partial\Psi}{\partial\alpha}\right)\left[2+\frac{(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})^2\varpi^2}{c^2}\left(1-\frac{\Sigma_{\mathrm{H}}a^2\sin^2\theta}{\rho_{\mathrm{H}}^4}\right)-4\frac{\Omega_{\mathrm{H}}(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})\varpi^2}{c^2}\left(1+\frac{\Sigma_{\mathrm{H}}(r_{\mathrm{H}}^2-a^2)}{2\rho_{\mathrm{H}}^2}\right)\right]+O(\alpha^2)$$

[see (6.7) in Macdonald & Thorne (1982)]. Equation (5.4) can of course be derived directly from the stream equation (2.13) in the same coordinate system. If Ψ is finite at the horizon, then the charge conservation law requires that

$$\Psi(\alpha,\lambda)\simeq\Psi_{\mathrm{H}}^{(0)}(\lambda)+\Psi_{\mathrm{H}}^{(1)}(\lambda)\alpha^2\ln\alpha+\Psi_{\mathrm{H}}^{(2)}(\lambda)\alpha^2+\cdots. \tag{5.5}$$

If, moreover, Ψ and I are well behaved on the axis of symmetry, then the condition (5.4) says that the absence of the $\alpha^2\ln\alpha$ term from Ψ, i.e. $\Psi_{\mathrm{H}}^{(1)}\equiv0$, is equivalent to the demand that

$$I(\Psi)=\pm(1/4\pi)\varpi_{\mathrm{H}}(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})d\Psi_{\mathrm{H}}^{(0)}/d\lambda=\pm(1/2)(\Omega_{\mathrm{H}}-\Omega_{\mathrm{F}})(B_{\mathrm{n}}\varpi^2)_{\mathrm{H}}, \tag{5.6a}$$

where one has, from (5.2a) and (5.5),

$$(B_{\mathrm{n}}\varpi^2)_{\mathrm{H}}=(1/2\pi)\varpi_{\mathrm{H}}d\Psi_{\mathrm{H}}^{(0)}(\lambda)/d\lambda. \tag{5.6b}$$

Just as one has chosen the sign $(+$ versus $-)$ in (4.7a) for I near infinity, so also one must choose the positive sign in (5.6a). Znajek pointed out (Macdonald & Thorne 1982) that the choice of sign $(+$ versus $-)$ corresponds to a choice of the direction of the Poynting flux at the horizon (in versus out) and hence to a choice of whether the horizon is a future horizon (black hole) or a past horizon (white hole). One can extend his suggestion to the situation at the infinity horizon: The choice of sign $(+$ versus $-)$

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in (4.7a) corresponds to a choice of the direction of the Poynting flux there (out versus in). Then the choice of the $-$ sign at both the stretched horizon and the infinity horizon implies the inverse process that the $e^{\mp}$-particles outflowing from a white hole meet with the $e^{\pm}$-particles ingoing from infinity to annihilate near the null surface.

If, then, $\Psi(\alpha, \lambda)$ behaves as given in (5.5) with $\Psi_H^{(1)}(\lambda) \equiv 0$, A in (A2a,b), B in (A3a,b), and the whole of the right-hand side of (5.4) are of the order of α^2. For the asymptotic values of electromagnetic fields, one has, from (5.2a,b), (5.6a,b) and (A4);

$$B_\theta \simeq -\frac{1}{2\pi c}\frac{g_H}{c^2}[2\alpha\Psi_H^{(2)}(\lambda)], \qquad E_r \simeq \frac{(\Omega_H-\Omega_F)}{2\pi c}\frac{g_H}{c^2}[2\Psi_H^{(2)}(\lambda)], \qquad E_\theta \simeq \frac{(\Omega_H-\Omega_F)}{2\pi c\alpha}\frac{d\Psi_H^{(0)}}{d\lambda}, \qquad j_T \simeq O(1). \tag{5.7a,b,c,d}$$

In order to obtain an explicit expression for j_T, one needs more careful calculation of all the terms of order α^2. It can be seen in (2.10), (5.3a,b), (5.6b) and (5.7a,b,c,d) that $E_\theta \simeq B_T \simeq \varrho_e \simeq |\boldsymbol{j}_P| \simeq O(\alpha^{-1})$, $B_r \simeq E_r \simeq j_T \simeq O(1)$ and $B_\theta \simeq O(\alpha)$ in the order of magnitude.

One can now see what the condition (5.6a) leads to. From (5.3a,b) and (5.6a) one obtains

$$\varrho_e \simeq \frac{(B_r)_H}{c\alpha}\frac{dI}{d\Psi}, \qquad \boldsymbol{j}_P \simeq c\varrho_e \boldsymbol{n}, \tag{5.8a,b}$$

which indicate that, if the plasma is charge-separated near the horizon with its velocity approaching the velocity of light, the positrons are blown down toward the horizon along the field lines in $\Psi_0 < \Psi < \Psi_c$ (Domain-$I^{(+)}$) and electrons are also blown down along the field lines in $\Psi_c < \Psi < \Psi_{eq}$ (Domain-$II^{(-)}$). Also, the condition (5.6a) combines with (2.7b) and (5.7c) to yield

$$\boldsymbol{B}_T \simeq -\frac{(\Omega_H-\Omega_F)}{2\pi c\alpha}\frac{d\Psi_H^{(0)}(\lambda)}{d\lambda}\boldsymbol{e}_{\hat{\phi}} \simeq -\boldsymbol{n}\times\boldsymbol{E}_\theta, \tag{5.9}$$

which is the radiative boundary condition showing that the tangential electric and magnetic fields look to ZAMOs like radiation propagating into the stretched horizon (Macdonald & Thorne 1982; Thorne *et al.* 1986). While the Poynting flux points outward and dominates the energy flux near the infinity horizon in (2.18a), that near the stretched horizon points inward, but the total flux $\boldsymbol{S}_E^P$ points outward, because the second term $\omega\boldsymbol{S}_J^P$ is larger than the Poynting flux in absolute value.

6 THE MOTION OF MASSLESS PLASMA IN THE FORCE-FREE REGIONS

The plasma in the force-free region is regarded as consisting of inertia-free particles, just introduced as charge carriers. Hence the motion of such a plasma can by no means be prescribed by the momentum equation, but must be evaluated via the electrodynamical quantities related to plasma motion. We presume that the plasma in the present context is the $e^{\pm}$-plasma due to their creation, and then relative number density must be evaluated from charge density ϱ_e and the difference of velocities weighted by number densities from the electric current $\boldsymbol{j}$:

$$n_+ - n_- = \varrho_e/e, \qquad n_+\boldsymbol{v}_+ - n_-\boldsymbol{v}_- = \boldsymbol{j}/e. \tag{6.1a,b}$$

In the force-free region one has eight unknowns for the four known quantities ϱ_e and $\boldsymbol{j}$, and therefore one cannot determine all of the unknowns uniquely in the domain of mixed plasma.

It was shown in the previous two sections that in the boundary regions of $\varpi \simeq 1/\alpha \simeq \infty$ the plasma is completely charge-separated, with velocity approaching the light velocity, that is, $|\boldsymbol{j}_P| \simeq c|\varrho_e|$, and that at either end of a given field line the sign of charge densities is different, thereby keeping the poloidal electric current along it in the same direction. This fact indicates that midway along given field lines emanating from the stretched horizon there must be a neutral domain where $\varrho_e \simeq 0$ and hence $n_+ \simeq n_- \simeq N/2$, where $n_+ + n_- = N$, and it will be reasonable to assume the domain containing the null surface of $\omega = \Omega_F$ and $\boldsymbol{v}_F = 0$ as defined by (3.7). In this quasi-neutral domain one has, from (2.10d) and (6.1d),

$$\boldsymbol{j}_P = -\frac{1}{\alpha}\frac{dI}{d\Psi} \simeq \frac{Ne}{2}(\boldsymbol{v}_+ - \boldsymbol{v}_-)_P. \tag{6.2}$$

Since $dI/d\Psi > 0$ in the region I where $\Psi_0 < \Psi < \Psi_c$, one sees $\boldsymbol{j}_P \propto (\boldsymbol{v}_+ - \boldsymbol{v}_-)_P < 0$, which is satisfied with $\boldsymbol{v}_{+P} < 0$ and $\boldsymbol{v}_{-P} > 0$, and thus the wind in the quasi-neutral domain of Region I consists of electron outflow and positron inflow. On the other hand, in Region II of $\Psi_c < \Psi < \Psi_{eq}$ where $dI/d\Psi < 0$, one finds that $\boldsymbol{j}_P \propto (\boldsymbol{v}_+ - \boldsymbol{v}_-)_P > 0$, which is satisfied with $\boldsymbol{v}_{+P} > 0$ and $\boldsymbol{v}_{-P} < 0$, and the wind consists of electron inflow and positron outflow.

To further extend our analysis, let us make a simplifying assumption that $n_+ = 0$ in Domain-$I^{(-)}$ and -$II^{(-)}$ where ϱ_e is negative and $n_- = 0$ in Domain-$I^{(+)}$ and -$II^{(+)}$ where ϱ_e is positive (see Fig. 2). Then except for thin neutral domains with $\varrho_e \simeq 0$, the plasma velocity is given in terms of a convection current of charges, i.e.

$$\boldsymbol{v} = \frac{\boldsymbol{j}}{\varrho_e}, \qquad \boldsymbol{v}_P = -\frac{1}{\varrho_e\alpha}\frac{dI}{d\Psi}\boldsymbol{B}_P, \qquad v_T = \frac{\varpi(\Omega_F-\omega)}{\alpha} + \frac{dI^2/d\Psi}{\varrho_e\alpha^2\varpi}. \tag{6.3a,b,c}$$

It is obvious that even if $\boldsymbol{v}$ stands for the velocity of massless particles, it must be smaller than the velocity of light, and one can

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Table 1. The sign of charge density of a charge-separated plasma in each domain, where ϖ_{IL}, ϖ_N and ϖ_{OL} denote the axial distances of the ILS, the null surface and the OLS, respectively.

	$\varpi < \varpi_{IL}$	$\varpi_{IL} < \varpi < \varpi_N$	$\varpi_N < \varpi < \varpi_{OL}$	$\varpi_{OL} < \varpi$
Region-I: $\Psi_0 < \Psi < \Psi_c$	$0 < \dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\omega-\Omega_F)}{\alpha c}+1} < \varrho_e < \dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\omega-\Omega_F)}{\alpha c}-1}$	$0 < \dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{1+\dfrac{\varpi(\omega-\Omega_F)}{\alpha c}} < \varrho_e$	$\varrho_e < -\dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{1+\dfrac{\varpi(\Omega_F-\omega)}{\alpha c}} < 0$	$-\dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\Omega_F-\omega)}{\alpha c}-1} < \varrho_e < -\dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\Omega_F-\omega)}{\alpha c}+1} < 0$
Region-II: $\Psi_c < \Psi < \Psi_{eq}$	$\dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\omega-\Omega_F)}{\alpha c}-1} < \varrho_e < \dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\omega-\Omega_F)}{\alpha c}+1} < 0$	$\varrho_e < \dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{1+\dfrac{\varpi(\omega-\Omega_F)}{\alpha c}} < 0$	$0 < -\dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\Omega_F-\omega)}{\alpha c}+1} < \varrho_e$	$0 < \dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\Omega_F-\omega)}{\alpha c}+1} < \varrho_e < -\dfrac{\dfrac{1}{\alpha^2 c^2 \varpi}\dfrac{dI^2}{d\Psi}}{\dfrac{\varpi(\Omega_F-\omega)}{\alpha c}-1}$

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impose for example $|v_T/c| \leq 1$. One can translate this inequality into such inequalities for ϱ_e in each domain as shown in Table 1. It can be seen that the previous choice of the sign of ϱ_e is consistent with the sign of ϱ_e given in Table 1.

Both expressions for $\boldsymbol{v}$ and ϱ_e contain the two parameter functions $I(\Psi)$ and $\Omega_F(\Psi)$. The stream equation (2.13) also contains them in a highly non-linear way, and to solve the partial differential equations, one must specify these two parameter functions. Making use of the charge conservation equation equivalent in the force-free plasma to the stream equation, one has already obtained the conditions (4.7a) and (5.6a) for the two functions from the boundary conditions for the behaviours of Ψ near the stretched horizon and infinity. We shall show in the following that one can obtain the same conditions for $I(\Psi)$ from the current stellar wind theory.

The theoretical degree of approximation changes from 'force-free', through 'pressure-free' (cold), to 'finite-temperature' (Mestel 1968; Goldreich & Julian 1969, 1970; Michel 1969, 1973; Julian 1973; Scharlemann & Wagoner 1973; Okamoto 1974, 1975, 1978; Kennel, Fujimura & Okamoto 1983). The steady axisymmetric theory tells us that there are generally three critical conditions to be satisfied at the sonic, Alfvénic and magnetosonic points, which uniquely determine related parameters including the one corresponding to $I(\Psi)$. In the force-free approximation there are two critical points remaining, that is, the light surface and the magnetosonic point at infinity. In the black hole's force-free magnetosphere one wind blows outward and the other inward from the quasi-neutral plasma spring situated near the null surface of $\omega = \Omega_F$ and $\boldsymbol{v}_F = 0$. Both outgoing and ingoing winds have respective critical points, that is, the outer and inner light surfaces and the magnetosonic points which are very near infinity and the hole's horizon. Mathematically the limit of $1/\alpha \to \infty$ corresponds to that of $\varpi \to \infty$. The critical conditions at the two light surfaces are given by (2.14) or (3.6). From now on we work out the critical conditions at the magnetosonic points.

One has a quadratic equation for $\boldsymbol{v}_P$ from (2.10b) and (6.3b,c);

$$v_P^2 + 2v_P\left[\frac{B_P B_T}{B^2}\,\frac{\varpi(\Omega_F-\omega)}{\alpha}\right] - \frac{B_P^2}{B^2}\left[v^2 - \frac{\varpi^2(\Omega_F-\omega)^2}{\alpha^2}\right] = 0, \tag{6.4a}$$

the solutions of which are given by

$$v_P = \frac{B_P}{B}\left[-\frac{B_T}{B}\,\frac{\varpi(\Omega_F-\omega)}{\alpha} \pm \left(v^2 - \frac{c^2E^2}{B^2}\right)^{1/2}\right], \tag{6.4b}$$

where $v^2 = v_P^2 + v_T^2$ and $E^2 = B_P^2\varpi^2(\Omega_F-\omega)^2/\alpha^2c^2$. For v_P to be real, one has a condition everywhere that

$$c|E/B| \leq v \leq c. \tag{6.5}$$

One can regard the magnetosonic point at $\varpi = \infty$ as an X-type critical point, although the right half of X is hidden beyond the horizon at infinity. The square root in (6.4b) must vanish with $v \to c$ for $\varpi \to \infty$, yielding

$$B^2 \simeq B_T^2 = (2I/c\varpi^2)^2 \simeq E^2 = B_P^2(\Omega_F\varpi/c)^2 \simeq E_\theta^2, \tag{6.6a}$$

because $B_T^2 \gg B_P^2 \simeq B_r^2$, and consequently

$$I(\Psi) = \pm(1/2)\Omega_F(B_r\varpi^2)_L \tag{6.6b}$$

where label L denotes the boundary between the force-free region and the load region beyond infinity. The result (6.6b) is nothing but the condition (4.7a) derived from the charge conservation law at the infinity horizon. The result (4.13) indicates that condition (6.6a) with $v \to c$ is equivalent to an 'outgoing-wave' boundary condition. The remaining part of (6.4b) at the magnetosonic point becomes

$$v_P/c \simeq (B_P/B)(-B_T/B)(\Omega_F\varpi/c) \simeq 1, \tag{6.7a}$$

and then from (6.3b,c), one obtains

$$\varrho_e \simeq -(B_r/c)(dI/d\Psi), \qquad v_T/c \simeq 0. \tag{6.7b,c}$$

It can easily be seen that (6.7a,c) is consistent with our presumption of $v \to c$, and (6.7b) coincides with (4.12c).

Quite similarly one can regard the magnetosonic point at the stretched horizon where $\omega \simeq \Omega_H$ for $\alpha \to 0$ as another X-type critical point for an ingoing centrifugal wind; in this case the left half of X is invisible because it is inside the stretched horizon. Requiring the square root in (6.4b) to vanish with $v \to c$ for $\alpha \to 0$, one has

$$B^2 \simeq B_T^2 = (-2I/c\alpha\varpi)^2 \simeq E^2 \simeq B_P^2[\varpi(\Omega_H-\Omega_F)/\alpha c]^2 \simeq E_\theta^2, \tag{6.8a}$$

and then

$$I(\Psi) = \pm(1/2)(\Omega_H-\Omega_F)(B_P\varpi^2)_H \tag{6.8b}$$

(see Blandford & Znajek 1977; Macdonald & Thorne 1982; Thorne *et al.* 1986). It can be seen that the critical condition at the magnetosonic point in the force-free approximation is equivalent to the 'ingoing-wave' boundary condition (5.9), and (6.8b) coincides with the result (5.6a) obtained from the charge conservation law at the stretched horizon. From the remaining part of (6.4b) and (6.3b,c) one obtains

$$v_P/c \simeq -(B_P\varpi^2)(\Omega_H-\Omega_F)/(2I) \simeq -1, \qquad \varrho_e \simeq (B_r/\alpha c)(dI/d\Psi), \qquad v_T/c \simeq 0. \tag{6.9a,b,c}$$

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These results given in (6.9a,b,c) agree with (5.8a,b). Also one can see that $v_T/c \simeq O(\alpha)$ for $\alpha \to 0$, since $j_T \simeq O(1)$ in (5.7d) and $\varrho_e \simeq O(\alpha^{-1})$ in (5.8a).

The convection current $\boldsymbol{j}$ is measured by the ZAMOs using their proper time and the convection velocity $\boldsymbol{v} = \boldsymbol{j}/\varrho_e$ is d(proper distance)$/d\tau$ relative to the ZAMOs. The velocity of $e^{\pm}$-particles on the global time, relative to absolute space, is

$$\frac{d(\text{proper distance})}{dt} = \alpha \boldsymbol{v} + \omega \boldsymbol{m} \tag{6.10}$$

(Macdonald & Thorne 1982). Near the horizon it becomes

$$\alpha \boldsymbol{v} + \omega \boldsymbol{m} \simeq \Omega_H \boldsymbol{m}, \tag{6.11}$$

which indicates that $e^{\pm}$-particles never cross the true horizon in any finite global time but orbit it forever. This leads to a boundary layer of complicated particle–field structures very near the true horizon, which Thorne *et al.* (1986) hide by stretching the horizon by a quantity α_H.

Making use of relativistic degenerate cold or warm stellar wind theory (Michel 1969; Goldreich & Julian 1970; Okamoto 1978), one can really confirm that the condition of the quadratic equation for v_P [given in (6.4a)] having double roots at infinity yields the critical condition at the magnetosonic point in the force-free limit. When one introduces dimensionless parameters

$$\lambda \equiv 2I/\Omega_F (B_P \varpi)_\infty, \qquad \sigma \equiv \Omega_F^2 (B_P \varpi^2)_\infty / 4\pi c^3 (\rho v_P \varpi^2)_\infty, \tag{6.12a,b}$$

the critical condition yields

$$u_\infty \equiv (\gamma v_P/c)_\infty = \sigma^{1/3}, \qquad \mu + \sigma\lambda = (1+\sigma^{2/3})^{3/2}, \tag{6.13a,b}$$

where μ is a constant along each field line, emanating from the energy integral of the momentum equation parallel to the poloidal field line. Condition (6.13a,b) holds for a general field structure of cold wind (Okamoto 1978). In the inertia-free limit of $\sigma \to \infty$, one finds $v_P \to c$ and $\lambda \to 1$, reproducing the result shown in (6.6b) and the presumption of $v \to c$. Directly requiring the denominator of the expression for the gradient of the Lorentz factor along each field line to vanish [see (2.18a) and (2.19a) in Okamoto 1978], one can obtain the result $B^2 \simeq E^2$ in the force-free limit, which is equivalent to the square root of (6.4b) being null. If one constructs a sophisticated theory for a non-force-free cold or hot ingoing wind due to the $\alpha\omega$-mechanism, or simply replaces Ω_F with $(\Omega_H - \Omega_F)$ in (6.12a,b), one will obtain the results given in (6.8a,b) in the massless limit. If one takes into account the correspondence to the wind theories in e.g., Okamoto (1978), it will turn out that the solution with a + sign in (6.4b) is an unphysical solution, while the one with a − sign is a physical solution. One should also choose (+ versus −) sign in expressions (6.6b) and (6.8b) for a black hole.

One cannot store energy and angular momentum in any part of the force-free region, and hence the inputs of energy and angular momentum per unit flux tube at the stretched horizon from the black hole to the force-free region must equal the outputs from the force-free region to the load region beyond infinity. Because $I(\Psi)/2\pi c$ and $\Omega_F(\Psi) I(\Psi)/2\pi c$ stand for the outward transport of angular momentum and energy per unit flux tube [see (2.18a,b)], one has enough reason to match the two conditions (6.6b) and (6.8b) for the two unknown functions of Ψ. Equating them yields

$$\Omega_F(\Psi) = \frac{\Omega_H}{1+\zeta}, \qquad I(\Psi) = \frac{1}{2}\frac{\zeta}{1+\zeta}\Omega_H (B_P \varpi^2)_H, \qquad \zeta(\Psi) \equiv \frac{(B_P \varpi^2)_L}{(B_P \varpi^2)_H}. \tag{6.14a,b,c}$$

Then from (2.18a,b) and (6.8b) with (+) sign one obtains;

$$\alpha \boldsymbol{S}_E^P = \frac{1}{4\pi c}\Omega_F(\Omega_H - \Omega_F)(B_P \varpi^2)_H \boldsymbol{B}_P = \frac{1}{4\pi c}\frac{\zeta}{(1+\zeta)^2}\Omega_H^2 (B_P \varpi^2)_H \boldsymbol{B}_P, \tag{6.15a,b}$$

$$\alpha \boldsymbol{S}_J^P = \frac{1}{4\pi c}(\Omega_H - \Omega_F)(B_P \varpi^2)_H \boldsymbol{B}_P = \frac{1}{4\pi c}\frac{\zeta}{1+\zeta}\Omega_H^2 (B_P \varpi^2)_H \boldsymbol{B}_P.$$

One can see that some sort of energy–angular-momentum relation, i.e. $\boldsymbol{S}_E^P = \Omega_F \boldsymbol{S}_J^P$ must hold everywhere in the force-free region. This is a result enforced by degeneracy (Macdonald & Thorne 1982). It can be seen in (6.15a) that $\zeta = 1$, i.e. $\Omega_F = \Omega_H/2$ optimizes the flux of redshift energy as measured using the global time (Blandford & Znajek 1977; Macdonald & Thorne 1982). It is thus quite significant to know the behaviour of $\zeta(\Psi)$. Expressions (6.14a,b,c) indicate that if the distributions of field lines at both ends are given from the boundary conditions at the two horizons, one can have ζ and then, Ω_F and I as functions of Ψ. As seen in Sections 4 and 5, if one should reject singularities near the two horizons, one has to assume the radial field distributions there, but there still remain ambiguities as to what forms are to be taken for functions $\Psi_L^{(0)}(\theta)$ in (4.6) and $\Psi_H^{(0)}(\lambda)$ in (5.5). One can regard $\zeta(\Psi)$ as a parameter indicating the degree of collimation of field lines in the force-free region.

In the black hole driven hydromagnetic flows, Phinney (1982) replaced the horizon boundary condition of the inflow effectively with the force-free condition (6.8b), but took into account the critical condition of a finite inertia cold wind for the outflow, which is equivalent to (6.13b). He obtained $\zeta \simeq 4$ for his field structure confined by the accretion disc and then $\Omega_F \simeq (1/5)\Omega_H$, using effectively the force-free condition even for the outflow. The case of $\zeta = 4$ in (6.15a) corresponds to an extracted power of 64 per cent of the case of $\zeta = 1$. Macdonald (1984) solved the stream equation (2.13) by a numerical method in the

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region of $r_H \le r \le 5r_H$, but since he imposed the condition that all of the field lines threading the horizon rotate with angular velocity $\Omega_F = \Omega_H/2$, i.e. $\zeta \equiv 1$, one can obtain no suggestions about $\zeta(\Psi)$ from his numerical models.

We have presumed a force-free magnetosphere surrounding a Kerr black hole and the associated accretion disc, in which all the information about the inertia of the particles is taken to infinity, and energy and angular momentum are transferred from electromagnetic to kinetic form through the Joule heating of the surface electric current on the 'sphere at infinity'. Although the present treatment is less strict than Phinney (1983), Camenzind (1989) and Takahasi *et al.* (1990), we adhere in this paper to the force-free approximation for the purpose of clarifying the magnetospheric physics and the evolution due to electrodynamical extraction of the hole's rotational energy.

7 STRUCTURE OF FORCE-FREE MAGNETOSPHERES

For a complete description of force-free magnetospheres, one need give $(B_P\varpi^2)_H$ and $(B_P\varpi^2)_L$ as the boundary conditions on the two horizons. Our present aim is to clarify the basic properties of the magnetospheric structure, and then we treat here the collimation parameter ζ as if it were constant from field line to field line.

7.1 Mid-surface

The ZAMOs' angular velocity ω decreases from Ω_H at the horizon surface to zero at infinity, and hence there is always (independently of ζ) a surface, the mid-surface (say) where $\omega = (1/2)\Omega_H$. If $\zeta = 1$, the null surface coincides with the mid-surface.

We introduce the dimensionless parameter $h \equiv a/r_H$ as indicating the rotational state. This can be shown to be significant for understanding the thermodynamical as well as evolutional states of Kerr holes (see OK-I, and OK-II). The h ranges from zero for the Schwarzschild state with $a = 0$ to unity for the extreme Kerr state with $a = r_H = GM_H/c^2$. Then, defining $x \equiv r/r_H$, one has from (2.1), (2.2) and (2.3),

$$F_M(x, \theta, h) \equiv (x^2 + h^2)(x^2 + h^2\cos^2\theta) - (1 + h^2)[2 + h^2(1 + \cos^2\theta)]x = 0, \tag{7.1}$$

for the equation determining the location of the mid-surface. It turns out in (7.1) that the location of the mid-surface x_M depends on h only. For slowly rotating holes with $h \ll 1$, one has

$$x_M \simeq 2^{1/3}\{1 + (h^2/6)[2(2 - 2^{1/3}) + (2^{1/3} - 1)\sin^2\theta]\}, \tag{7.2a}$$

and for rapidly rotating holes with $\varepsilon \equiv 1 - h^2 \ll 1$, (7.1) reduces to

$$(x^2 + 1)^2 - 8x - (x - 1)^2\sin^2\theta - \varepsilon[2(x^2 - 4x + 1) - (x - 1)(x - 2)\sin^2\theta] = 0. \tag{7.2b}$$

For the limit of $h \to 0$, one sees that the mid-surface tends to a constant radius, i.e. $x_M \to 2^{1/3} = 1.2599$. For $h^2 = 2^{1/2} - 1$, one obtains $x_M = 1.3960$ at $\theta = 0$ and $x_M = 1 + h^2 = 2^{1/2}$ at $\theta = \pi/2$. In the limit of $h = 1$, one has $x_M = 1.6085$ at $\theta = 0$ and $x_M = 1.6344$ at $\theta = \pi/2$ from (7.2b). One thus sees that $2^{1/3} \le x_M \le 1.6344$ for whole ranges of h and θ.

7.2 Null surface

The null surface is defined by (3.7), and using (2.1–2.3), one can transform it into an algebraic form

$$F_N(x, \theta, h, \zeta) \equiv (x^2 + h^2)(x^2 + h^2\cos^2\theta) - (1 + h^2)[1 + h^2\cos^2\theta + \zeta(1 + h^2)]x = 0. \tag{7.3}$$

Then, depending upon ζ, one has approximately

$$x_N \simeq \begin{cases} 1, & \text{for } \zeta \simeq 0, \\ x_M, & \text{for } \zeta \simeq 1, \\ \zeta^{1/3}(1 + h^2)^{2/3}, & \text{for } \zeta \gg 1. \end{cases} \tag{7.4}$$

7.3 Inner and outer light surfaces

For later reference we define the equation of the static-limit surface, i.e. $g_{tt} = -(\Delta - a^2\sin^2\theta)/\rho^2 = 0$, as follows:

$$F_E(x, \theta, h) \equiv (x - 1)(x - h^2) - h^2\sin^2\theta, \tag{7.5a}$$

and the radius of the surface is given by

$$x_E = \frac{1 + h^2}{2}\left[1 + \left\{1 - \left(\frac{2h\cos\theta}{1 + h^2}\right)^2\right\}^{1/2}\right]. \tag{7.5b}$$

The inner and outer light surfaces are defined by (2.14b). These are expressed in the algebraic form by

$$F_{LS}(x, \theta, h, \zeta) \equiv \frac{1}{1 + \zeta}[(x^2 + h^2)^2 - h^2\sin^2\theta(x - 1)(x - h^2)] - (1 + h^2)^2 x \pm (1 + \zeta)(1 + h^2)\frac{x^2 + h^2}{h\sin\theta}(x - 1)^{1/2}(x - h^2)^{1/2} = 0, \tag{7.6}$$

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where the upper sign is for the inner light surface (ILS) and the lower sign is for the outer light surface (OLS). In the polar region with $\cos^2\theta \simeq 1$ for the case of $h \neq 1$, one has, from (7.6),

$$x_{\rm IL} \simeq 1 + \frac{\zeta^2 h^2 \sin^2\theta}{(1+\zeta)^2(1-h^2)}, \qquad x_{\rm OL}\sin\theta \simeq (1+h^2)\left[\frac{1+\zeta}{h} - \frac{\sin\theta}{2}\right]. \tag{7.7a,b}$$

Thus, in the polar region, the ILS approaches the horizon like $\sin^2\theta$, and the OLS asymptotically tends to a 'cylinder' with radius $x_{\rm OL}\sin\theta = (1+\zeta)(1+h^2)/h$. It can also be seen in (7.7a,b) that for $\zeta \to 0$, $x_{\rm IL} \to 1$ and $x_{\rm OL}\sin\theta \simeq (1+h^2)[1/h - \sin\theta/2]$, while for $\zeta \gg 1$, $x_{\rm IL} \simeq 1 + [h^2/(1-h^2)]\sin^2\theta$, and $x_{\rm OLS}\sin\theta \simeq (1+h^2)[\zeta/h - \sin\theta/2]$. For the limit of $h \to 0$, the ILS falls down on to the horizon, and the OLS disappears beyond infinity. In the equatorial region of a slowly rotating hole, one has, from (7.6),

$$x_{\rm IL} \simeq 1 + \frac{\zeta^2 h^2}{(1+\zeta)^2}, \qquad x_{\rm OL}\sin\theta \simeq \frac{1+\zeta}{h} - \frac{1}{2}. \tag{7.8a,b}$$

For the maximally rotating holes with $h = 1$, (7.6) reduces to

$$F_{\rm LS}(x, \theta, 1, \zeta) = \frac{1}{1+\zeta}[(x^2+1)^2 - (x-1)^2\sin^2\theta] - 4x \pm 2\left[\frac{x^2+\cos^2\theta}{\sin\theta}\right](x-1). \tag{7.9}$$

Then, one obtains

$$x_{\rm IL} \simeq 1 + \frac{\zeta}{1+\zeta}\sin\theta, \qquad x_{\rm OL}\sin\theta \simeq 2(1+\zeta) - \sin\theta \tag{7.10a,b}$$

in the polar region, and

$$x_{\rm IL} = (\zeta^2 + 5\zeta + 9/4)^{1/2} - (\zeta + 1/2), \qquad x_{\rm OL} = 2\zeta + 1 \tag{7.11a,b}$$

in the equatorial region.

The expressions (7.7a), (7.8a), (7.10a) and (7.11a) for $x_{\rm IL}$ in the case of $\zeta \to \infty$ coincide with (7.5b). One can in general show that the ILS for $\zeta \to \infty$, i.e. $\Omega_{\rm F} \to 0$, reduces to the static-limit surface; one has $\alpha \simeq \omega\varpi/c$ from (2.14b) for $\Omega_{\rm F} \simeq 0$, and using (2.1), one obtains a relation

$$\alpha^2 - (\omega\varpi/c)^2 = (\Delta - a^2\sin^2\theta)/\rho^2 = 0, \tag{7.12}$$

of which solution is given by (7.5b). Thus, the outermost location of the ILS is given by the static-limit surface, while the innermost location of the OLS is $x_{\rm OL}\sin\theta = 1$ for $\theta = \pi/2$ and $x_{\rm OL}\sin\theta = 2 - \sin\theta$ for $\theta \simeq 0$ in the case of $h = 1$ and $\zeta = 0$, i.e. $\Omega_{\rm F} = \Omega_{\rm H}$.

7.4 'Effective ergosphere'

The usual ergosphere is defined by the static-limit surface, expressed in our notations by (7.5a,b). Although the 'effective ergosphere' is defined as the region for a charged black hole in which there exist orbits of negative energy-at-infinity for charged particle processes (Misner *et al.* 1973), we here refer to the region with $\varepsilon_{\rm E} < 0$ as the same 'effective ergoregion'. From (2.17a,b) and (6.14a,c) one obtains

$$\varepsilon_{\rm E} = \frac{B_{\rm P}^2}{8\pi\alpha^2}\left[\alpha^2 + \frac{\Omega_{\rm H}^2\varpi^2}{c^2(1+\zeta)^2}\left\{1 + \zeta^2\frac{(B_{\rm P}\varpi^2)_{\rm H}^2}{(B_{\rm P}\varpi^2)^2}\right\} - \frac{\omega^2\varpi^2}{c^2}\right], \qquad \varepsilon_{\rm J} = \frac{B_{\rm P}\varpi^2}{4\pi\alpha c^2}\left[\frac{\Omega_{\rm H}}{1+\zeta} - \omega\right]B_{\rm P}. \tag{7.13a,b}$$

In the limit of $\zeta \to 0$, (7.13a,b) reduce to

$$\varepsilon_{\rm E} \simeq \frac{B_{\rm P}^2}{8\pi}\left[1 + \frac{(\Omega_{\rm H}^2 - \omega^2)\varpi^2}{\alpha^2 c^2}\right] > 0, \qquad \varepsilon_{\rm J} \simeq \frac{(\Omega_{\rm H} - \omega)(B_{\rm P}\varpi^2)}{4\pi\alpha c^2}B_{\rm P} \geq 0. \tag{7.14a,b}$$

Although the region of negative $\varepsilon_{\rm J}$ does not coincide with that of negative $\varepsilon_{\rm E}$, both the regions shrink toward the horizon, as ζ becomes smaller and smaller, and vanish in the limit of $\zeta \to 0$. The value of $\varepsilon_{\rm E}$ remains finite and that of $\varepsilon_{\rm J}$ becomes null on the horizon.

Next let us consider the opposite limit of $\zeta \to \infty$, assuming that $B_{\rm P}\varpi^2 \gg (B_{\rm P}\varpi^2)_{\rm H}$ in the vicinity of the surface of $\varepsilon_{\rm H} = 0$. Then, (7.13a,b) reduce to

$$\varepsilon_{\rm E} \simeq \frac{B_{\rm P}^2}{8\pi\alpha^2}\left[\alpha^2 - \left(\frac{\omega\varpi}{c}\right)^2\right], \qquad \varepsilon_{\rm J} \simeq -\frac{\omega B_{\rm P}\varpi^2}{4\pi\alpha c^2}B_{\rm P}. \tag{7.15a,b}$$

Comparing (7.15a) with (7.12), one sees that the effective ergosphere reduces to the static-limit surface for $\zeta \to \infty$, although $\varepsilon_{\rm J}$

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seems to be everywhere negative. For later convenience, we express (7.12) in the form

$$F_{EE}(x,\theta,h,\infty)\equiv(x-1)(x-h^2)-h^2\sin^2\theta\left[\frac{(1+h^2)x}{x^2+h^2\cos^2\theta}\right]^2=\frac{\Sigma^2}{\rho^4}F_E(x,\theta,h)=0, \tag{7.16}$$

where $F_E(x,\theta,h)$ is given by (7.5a).

For intermediate values of ζ, one has from (2.1), (2.12a), (7.13a) and (7.16)

$$F_{EE}(x,\theta,h,\zeta)\equiv F_{EE}(x,\theta,h,\infty)+\left[\frac{h^2\sin^2\theta}{(1+h^2)^2}\right]\left[\frac{(1+\zeta)^2[(B_P\varpi^2)_H^2/(B_P\varpi^2)^2]}{1+\zeta^2}\right]\frac{\Sigma^4}{\rho^8}(x^2+h^2\cos^2\theta)^2 \tag{7.17}$$

$$=\frac{\Sigma^2}{\rho^4}\left\{F_E(x,\theta,h)+\left[\frac{h^2\sin^2\theta}{(1+h^2)^2}\right]\left[\frac{1+\zeta^2[(B_P\varpi^2)_H^2/(B_P\varpi^2)^2]}{(1+\zeta)^2}\right]\frac{\Sigma^2}{\rho^4}(x^2+h^2\cos^2\theta)^2\right\}=0.$$

This is the equation determining the location of the surface of the effective ergosphere for finite ζ. Substituting $x=x_E$ from (7.5b) and $x=1$ into (7.17), one sees that

$$F_{EE}(x_E,\theta,h,\zeta)>0,\qquad F_{EE}(1,\theta,h,\zeta)=-h^2\sin^2\theta\frac{(1+h^2)^2}{(1+h^2\cos^2\theta)^2}\frac{2\zeta}{(1+\zeta)^2}<0, \tag{7.18a,b}$$

which imply that the effective ergosphere for finite ζ must always lie inside the usual ergosphere in the force-free case. In Fig. 3 is depicted the effective region for the case of $h=1$ and $\zeta=1$, together with the static-limit surface and the null mid-surface.

7.5 Critical field line

The definition of the critical field line (CFL) is given in (3.3) and (3.4); it is the field line that maximizes the function of $I(\Psi)$. Then one can expand it in the vicinity of the CFL;

$$I(\Psi)=I_c+(1/2)(d^2I/d\Psi^2)_c(\Psi-\Psi_c)^2+\ldots \tag{7.19}$$

where $I_c=I(\Psi_c)$, and $(d^2I/d\Psi^2)_c<0$. The CFL is completely charge-separating at least near the stretched horizon and the infinity horizon, and hence $\varrho_e\propto-(\Psi-\Psi_c)$ for $\alpha\to0$ from (5.8a) and $\varrho_e\propto+(\Psi-\Psi_c)$ for $\omega\to\infty$ from (4.12c), in the vicinity of the CFL. We have assumed in Section 6 that Ψ_c separates the region of $\varrho_e<0$ from that of $\varrho_e>0$ all the way from the stretched horizon to the infinity horizon except for the null domains $\mathrm{I}^{(0)}$ and $\mathrm{II}^{(0)}$; if we continue to assume so, from (6.3b) and (7.19) one has, in the neighbourhood of Ψ_c,

$$\boldsymbol{v}_P\simeq-(\varrho_e\alpha)^{-1}(d^2I/d\Psi^2)_c(\Psi-\Psi_c)\boldsymbol{B}_P, \tag{7.20}$$

which is valid across the CFL from the domains $\mathrm{I}^{(\pm)}$ to the domains $\mathrm{II}^{(\mp)}$ and $\boldsymbol{v}_P$ may remain finite at $\Psi=\Psi_c$ in both outgoing and

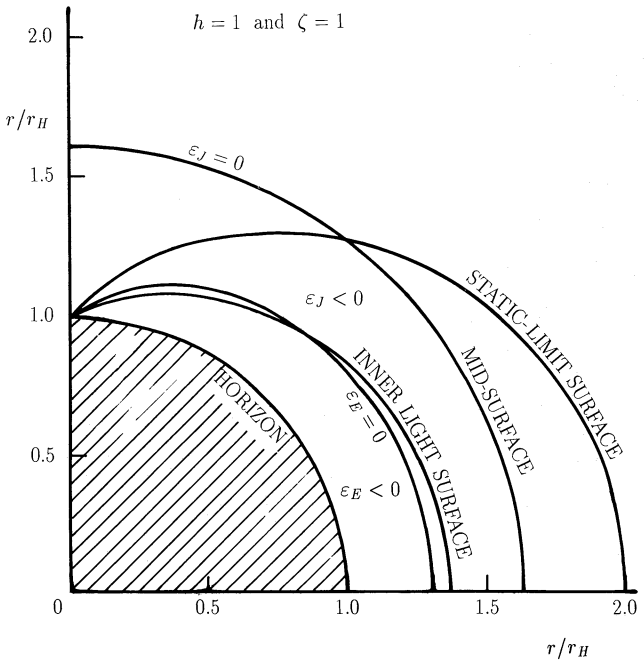


Figure 3. The 'effective' ergoregion in the case of $h=1$ and $\zeta=1$.

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ingoing flows. Thus along the CFL one has the following conditions, from (3.2a),

$$\frac{dI}{d\Psi} = \varrho_e = \nabla\Psi\cdot\boldsymbol{G} = 0. \tag{7.21}$$

Conditions (7.21) are reminiscent of the pulsar case in the flat space-time; in the steady axisymmetric theory of the force-free magnetosphere with charge-separated plasma, the CFL must be a straight line parallel to the equator owing to rigidity of the pulsar surface (Okamoto 1974). One can easily confirm in (4.2a) that $\boldsymbol{G}\cdot\nabla\Psi = 2\pi c^2 B_z = 0$ along the CFL in the case of $d\Omega_F/d\Psi = 0$. The ZAMOs living deep in the force-free magnetosphere, on the other hand, feel sufficiently strong dragging of the inertial frame. It is actually the $\alpha\omega$-mechanism that causes the inner and outer light surfaces to be curved from a cylindrical shape. This may similarly produce a sophisticated topology for the CFL which would look unreasonable in the pulsar case. From (3.1b) and (6.14b) one has

$$\boldsymbol{G} = \boldsymbol{g} - \frac{c^2}{2}\left[1 + \frac{\varpi^2[\Omega_H/(1+\zeta)-\omega]^2}{\alpha^2 c^2}\right]\frac{\nabla\omega}{\Omega_H/(1+\zeta)-\omega} + c^2\nabla\ln\varpi - \frac{c^2}{2}\left[\frac{\Omega_H}{\Omega_H/(1+\zeta)-\omega}\right]\frac{\zeta'}{(1+\zeta)^2}\nabla\Psi, \tag{7.22}$$

where $\zeta' = d\zeta/d\Psi$, and the first two terms are due to the $\alpha\omega$-mechanism. In the neighbourhood of the two horizons, (7.22) reduces to

$$\boldsymbol{G} \simeq \begin{cases} -\dfrac{\boldsymbol{n}}{\alpha}\left\{g_H + \dfrac{c^2}{2}\left(\dfrac{\Omega_H\varpi_H}{c}\dfrac{2GM_H}{c^2 r_H}\right)^2\left(\dfrac{\zeta}{1+\zeta}\right)\dfrac{r_H^3}{\rho_H^2\Sigma_H^3}[\Sigma_H(3r_H^2-a^2)-a^2(r_H^2-a^2)\sin^2\theta]\right\}, & \text{for } \alpha\to 0, \quad (7.23a) \\ c^2\left\{\dfrac{\boldsymbol{e}_\varpi}{\varpi} - \dfrac{1}{2}\dfrac{\zeta'}{1+\zeta}\nabla\Psi - \dfrac{GM_H}{c^2 r^2}\left[1 - 3\left(\dfrac{\Omega_H\varpi_H}{c}\dfrac{2GM_H}{c^2 r_H}\right)^2\dfrac{\sin^2\theta}{1+\zeta}\right]\boldsymbol{n}\right\}, & \text{for } \varpi\to\infty, \quad (7.23b) \end{cases}$$

where $\Sigma_H = r_H^2 + a^2$, $\rho_H^2 = r_H^2 + a^2\cos^2\theta$ and $\boldsymbol{e}_\varpi$ is the unit vector parallel to the ϖ-axis in cylindrical coordinates.

It can be seen in (7.23a) and (5.5) that near the stretched horizon where the $\alpha\omega$ effects are largest in the magnetosphere, $\boldsymbol{G}\propto\boldsymbol{n}$, $\nabla\Psi \simeq [d\Psi_H^{(0)}(\lambda)/d\lambda]\boldsymbol{e}_\theta$ and hence $\nabla\Psi\cdot\boldsymbol{G} = 0$ for every field line including the CFL, which signifies that the CFL can emanate normally from the stretched horizon, although this is obvious in (5.6b). After starting normally from there, the CFL passes through the ILS, the null surface, the mid-surface, and then the OLS. If the OLS is in the region of curved space-time, the first two terms will still prevail in addition to the last term in (7.22), so that the CFL need not be parallel to the equator. In the region of nearly flat space-time, (7.23b) indicates that the magnitude of the $\alpha\omega$-mechanism decreases like $O(r^{-2})$. If the space-time is flat in the neighbourhood of the boundary of the force-free region to the load region, one has $B_z = \pi[\zeta'/(1+\zeta)]B_P^2$ from $\boldsymbol{G}\cdot\nabla\Psi = 0$. However, the plasma velocity there will become so large that one has to include the plasma inertia, which induces the transfield motion and the charge mixing. Thus, the CFL will lose the role of the 'critical field line' separating the region of positive charge from that of negative charge.

8 ELECTROMAGNETIC PROCESSES ON THE TWO HORIZONS – MEMBRANES

The ZAMOs in the force-free region are destined to a narrow field of vision so that they can neither observe the hole's interior beyond the event horizon nor the outer world beyond the force-free region. Thorne *et al.* (1986) presumed the presence of a boundary layer with a suitably chosen quite small thickness α_H just above the true event horizon, and regarded the stretched horizon as a membrane (referred to here as 'H-membrane') with various physical properties. So also in the approximation of force-freeness one can presume the presence of another membrane (referred to as 'L-membrane') at the horizon of the load region. Just as the H-membrane performs for the black hole inside, so does the L-membrane for the load region outside (see Fig. 2). One can see in the following that the L-membrane has quite similar but antisymmetric properties to the H-membrane.

8.1 The H-membrane

Following Thorne *et al.*'s membrane paradigm, let us now consider the physical roles of the H-membrane. The membrane fields are defined as the ones multiplied by gravitational redshift factor α at the stretched horizon, then one has from (5.7c) and (2.10b);

$$\boldsymbol{E}_H \equiv (\alpha E_\theta)_H\boldsymbol{e}_\theta = [(\Omega_H - \Omega_F)/2\pi c][d\Psi_H^{(0)}/d\lambda]\boldsymbol{e}_\theta = [\zeta/(1+\zeta)](\Omega_H\varpi_H/c)B_r\boldsymbol{e}_\theta, \qquad \boldsymbol{B}_H \equiv (\alpha\boldsymbol{B}_T)_H = -(2I/c\varpi)\boldsymbol{e}_\phi = -|\boldsymbol{E}_H|\boldsymbol{e}_\phi. \tag{8.1a,b}$$

These are the electric and magnetic fields measured by ZAMOs, if the ZAMOs use global time t rather than proper time τ. The other components, if measured using global time t, are of order of α_H or smaller. As one has seen in (5.9), the radiative boundary condition

$$\boldsymbol{B}_H = \boldsymbol{E}_H \times \boldsymbol{n} \tag{8.2}$$

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holds on the H-membrane. The amount of charge falling down on to or flowing out of the H-membrane per unit area per unit global time is given from (5.8a, b) and (5.6b) by

$$(\alpha j_r)_{\rm H} = -c(\alpha \varrho_{\rm e})_{\rm H} = -(1/2\pi\varpi_{\rm H})[d\Psi_{\rm H}^{(0)}/d\lambda][dI/d\Psi] = -(B_r)_{\rm H}(dI/d\Psi). \tag{8.3}$$

The surface charge density is defined by Gauss' law; it is the charge density required to terminate the electric field lines which thread the stretched horizon,

$$\sigma_{\rm eH} \equiv (E_{\rm n}/4\pi)_{\rm H} = [(\Omega_{\rm H} - \Omega_{\rm F})/8\pi^2 c][g_{\rm H}/c^2][2\Psi_{\rm H}^{(2)}], \tag{8.4}$$

where $E_{\rm n}$ is the normal component of the electric field given by (5.7b). The total surface charge becomes, from (8.4) and (2.6a),

$$Q_{\rm H} = \oint \sigma_{\rm eH}\, dA_{\rm H} = (g_{\rm H}/2\pi c^3) \oint (\Omega_{\rm H} - \Omega_{\rm F})\, \varpi_{\rm H} \Psi_{\rm H}^{(2)}(\lambda)\, d\lambda, \tag{8.5}$$

which may not vanish, if the total surface charge on the L-membrane does not vanish [*cf.* (8.21)].

The surface current $\mathcal{J}_{\rm eH}$ is defined as charge crossing a unit length perpendicular to $\mathcal{J}_{\rm eH}$ on the stretched horizon with circular length $2\pi\varpi$, per unit global time; from the definition of $I(\Psi)$, one has

$$\mathcal{J}_{\rm eH} = [I(\Psi)/2\pi\varpi_{\rm H}]\, \boldsymbol{e}_{\hat{\theta}} = [(\Omega_{\rm H} - \Omega_{\rm F})/8\pi^2][d\Psi_{\rm H}^{(0)}/d\lambda]\, \boldsymbol{e}_{\hat{\theta}} = (c/4\pi)\boldsymbol{E}_{\rm H}, \tag{8.6}$$

where (5.6a) and (8.1a) are used. The surface current of course coincides with that defined by Ampère's law, that is, the current density required to terminate the tangential magnetic field $\boldsymbol{B}_{\rm H}$ on the stretched horizon

$$\boldsymbol{B}_{\rm H} = (4\pi/c)(\mathcal{J}_{\rm eH} \times \boldsymbol{n}). \tag{8.7}$$

Expression (8.6) indicates that Ohm's law holds on the H-membrane with the surface electric resistivity

$$R_{\rm H} = 4\pi/c. \tag{8.8}$$

It turns out in (8.4), (8.6) and (8.8) that the surface electric current on the H-membrane is conduction current and its velocity is $c(E_{\rm H}/E_{\rm n}) = c(c^2/g_{\rm H})[d\Psi_{\rm H}^{(0)}/d\lambda]/2\Psi_{\rm H}^{(2)}$. One can also see in the stationary state that expressions (8.3) and (8.6) satisfy the law of charge conservation on the H-membrane,

$$(\partial\sigma_{\rm eH}/\partial t) + {}^{(2)}\nabla \cdot \mathcal{J}_{\rm eH} + (\alpha j_r)_{\rm H} = 0, \tag{8.9}$$

where ${}^{(2)}\nabla \cdot \mathcal{J}_{\rm eH} = (1/\varpi_{\rm H})[\partial(\varpi \mathcal{J}_{{\rm eH},\theta})/\partial\lambda]$ is the two-dimensional divergence of the surface current on the H-membrane with axial symmetry.

The electromagnetic field produces the surface Lorentz force on a unit area of the H-membrane given by

$$\sigma_{\rm eH}\boldsymbol{E}_{\rm H} + (\mathcal{J}_{\rm eH}/c) \times \boldsymbol{B}_{\rm n} = -\frac{I(\Psi)}{2\pi c\varpi} B_r \boldsymbol{e}_{\hat{\phi}} + \frac{g_{\rm H}}{c^2}\left(\frac{\Omega_{\rm H} - \Omega_{\rm F}}{2\pi c}\right)^2 \frac{d\Psi_{\rm H}^{(0)}}{d\lambda}\frac{\Psi_{\rm H}^{(2)}}{2\pi}\, \boldsymbol{e}_{\hat{\theta}}, \tag{8.10}$$

where (8.1a), (8.4) and (8.6) are substituted. Just outside the stretched horizon, the condition of force-freeness, especially in the poloidal direction, is maintained by the non-zero toroidal current $j_{\rm T}$ coupled with charge density $\varrho_{\rm e}$ in spite of axial symmetry with $\boldsymbol{E}_{\rm T} \equiv 0$. On the H-membrane with electric resistivity is, on the other hand, no toroidal electric field and hence no toroidal conductive electric current present, and therefore the force balance in both the poloidal and toroidal directions is lost, thereby giving rise to a powerful electromotive force which maintains the system of electric circuit. This plays a significant role in extracting rotational energy and angular momentum from the black hole (Blandford & Znajek 1977; Macdonald & Thorne 1982; Phinney 1982; Thorne *et al.* 1986).

The torque per unit global time on the H-membrane is given by integrating the Lorentz force multiplied with the lever arm $\varpi \boldsymbol{e}_{\hat{\phi}}$. It is equal to minus the outward flux of angular momentum due to the electromagnetic field; from (8.10) and (2.18b) one has

$$dJ_{\rm H}/dA_{\rm H}\, dt = -\alpha \boldsymbol{S}_{\rm J} \cdot \boldsymbol{n} = -[I(\Psi)/2\pi c] B_r. \tag{8.11a}$$

As one can see in (2.17) or (7.13b), angular momentum density $\varepsilon_{\rm J}$ is negative inside the null surface, and tends near the H-membrane from (5.16b) to $\alpha\varepsilon_{\rm J} \simeq -[I/2\pi c](B_r/c)$. Then one finds that

$$\alpha \boldsymbol{S}_{\rm J} \cdot \boldsymbol{n} \simeq (-c)(\alpha\varepsilon_{\rm J}), \tag{8.11b}$$

which indicates that the ingoing wind convects negative angular momentum towards the H-membrane and consequently transports angular momentum out of it. Integrating (8.11a) over all the field lines emanating from the H-membrane, one obtains the rate of angular momentum loss from the hole,

$$dJ_{\rm H}/dt = -\oint \alpha \boldsymbol{S}_{\rm J} \cdot \boldsymbol{n}\, dA_{\rm H} = -(1/2\pi c)\oint I(\Psi)\, d\Psi. \tag{8.12}$$

The surface electric current driven by the surface electric field $\boldsymbol{E}_{\rm H}$ on the resistive membrane produces Joule heating, thereby increasing the hole's entropy (see e.g. Thorne *et al.* 1986; also OK-I). The rate of increase of entropy becomes, from (8.6) and

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(8.8),

$$T_{\rm H}(dS_{\rm H}/dA_{\rm H}dt)=\boldsymbol{E}_{\rm H}\cdot\boldsymbol{\mathcal{J}}_{\rm eH}=R_{\rm H}|\boldsymbol{\mathcal{J}}_{\rm eH}|^2=(1/2\pi c)(\Omega_{\rm H}-\Omega_{\rm F})\,I(\Psi)B_r, \tag{8.13}$$

where $T_{\rm H}$ and $S_{\rm H}$ are the hole's temperature and entropy, respectively, related to $g_{\rm H}$ and $A_{\rm H}$ as follows:

$$T_{\rm H}=(\hbar/2\pi k_{\rm B}c)g_{\rm H},\qquad S_{\rm H}=(c^3k_{\rm B}/4G\hbar)A_{\rm H} \tag{8.14a,b}$$

(Bekenstein 1975; Hawking 1976). Integrating (8.13) over all the field lines on the H-membrane yields

$$T_{\rm H}\frac{dS_{\rm H}}{dt}=\frac{1}{2\pi c}\oint(\Omega_{\rm H}-\Omega_{\rm F})\,I(\Psi)\,d\Psi. \tag{8.15}$$

If there is a non-zero total charge present on the stretched horizon, the space around a charged hole must be described by the Kerr–Newman metric, and the total mass $M_{\rm H}$ must contain a contribution from the total charge $Q_{\rm H}$, from (8.5). However, just as we presume no total charge on the L-membrane at the infinity horizon acting for the load region [see (8.21)], so we require $Q_{\rm H}\equiv 0$ on the H-membrane. Then, the first law of black hole thermodynamics for a rotating uncharged hole leads to the change of the total mass due to the electrodynamical extraction of energy; from (8.12) and (8.15) one has

$$c^2\frac{dM_{\rm H}}{dt}=T_{\rm H}\frac{dS_{\rm H}}{dt}+\Omega_{\rm H}\frac{dJ_{\rm H}}{dt}=-\frac{1}{2\pi c}\oint\Omega_{\rm F}(\Psi)\,I(\Psi)\,d\Psi=-\frac{1}{2c}\oint\Omega_{\rm F}(\Omega_{\rm H}-\Omega_{\rm F})\langle B_{\rm n}\varpi\rangle_{\rm H}^2\,d\lambda=-P_{\rm H}, \tag{8.16}$$

where $P_{\rm H}$ is the redshifted energy per global time flowing out of the H-membrane.

The power released from the H-membrane is equal to the total outward energy flux. The poloidal component of energy flux (2.18a) consists of the two terms, which reduce for $\alpha\to 0$ to

$$\alpha\boldsymbol{S}_{\rm E}^{\rm P}\to[(c/4\pi)(\boldsymbol{E}_{\rm H}\times\boldsymbol{B}_{\rm H})+\Omega_{\rm H}\langle\varpi\rangle_{\rm H}|\boldsymbol{B}_{\rm H}|/4\pi)\boldsymbol{B}_{\rm P}]=[-(\boldsymbol{E}_{\rm H}\cdot\boldsymbol{\mathcal{J}}_{\rm eH})+(\Omega_{\rm H}I/2\pi c)B_r]\boldsymbol{n}. \tag{8.17}$$

The first term is the Poynting flux pointing inward and contributes to the entropy production given in (8.13) or (8.15) and this reflects the situations where the electromagnetic field looks to a ZAMO near the horizon like an infinitely blueshifted, ingoing electromagnetic plane wave. The second term represents the outward removal of rotational energy corresponding to angular momentum loss multiplied by $\Omega_{\rm H}$. These two sum to yield the outward energy flux. Then, why can the ingoing wind transport energy out of the horizon? The answer is that just as it carries negative angular momentum into the hole, so it convects negative energy into the hole, that is, from (2.17a), (2.18a) and (5.6a), one has

$$\alpha\boldsymbol{S}_{\rm E}\cdot\boldsymbol{n}\simeq(-c)(\alpha\varepsilon_{\rm E})\simeq[\Omega_{\rm F}(\Psi)\,I(\Psi)/2\pi c]B_r. \tag{8.18}$$

Integrating (8.18) on the H-membrane yields the power $P_{\rm H}$ given in (8.16).

We have seen in Section 7 that $\zeta=1$, i.e. $\Omega_{\rm F}\simeq\Omega_{\rm H}/2$, optimizes the outward flux of redshifted energy from the hole [see (6.15a)]. One can easily confirm this fact in (8.16) for the total power. For the case of $\zeta\ll 1$, i.e. $\Omega_{\rm F}\simeq\Omega_{\rm H}$, it can be seen in (6.14b), (8.1a,b), (8.6), (8.12) and (8.15) that

$$I(\Psi)\simeq\zeta(\Omega_{\rm H}/2)\langle B_{\rm P}\varpi^2\rangle_{\rm H},\qquad |\boldsymbol{E}_{\rm H}|\simeq|\boldsymbol{B}_{\rm H}|\simeq(4\pi/c)|\boldsymbol{\mathcal{J}}_{\rm eH}|\simeq\zeta(\Omega_{\rm H}\langle\varpi\rangle_{\rm H}/c)B_r, \tag{8.19a}$$

which indicate that the membrane fields on the stretched horizon become weaker and weaker with ζ becoming small. On the other hand, for the case of $\zeta\gg 1$, i.e. $\Omega_{\rm F}\ll\Omega_{\rm H}$, one obtains

$$I(\Psi)\simeq(\Omega_{\rm H}/2)\langle B_{\rm P}\varpi^2\rangle_{\rm H},\qquad |\boldsymbol{E}_{\rm H}|\simeq|\boldsymbol{B}_{\rm H}|\simeq(4\pi/c)|\boldsymbol{\mathcal{J}}_{\rm eH}|\simeq(\Omega_{\rm H}\langle\varpi\rangle_{\rm H}/c)B_r, \tag{8.19b}$$

which implies that the membrane fields reduce to the values independent of ζ [*cf.* (8.31a,b)].

8.2 The L-membrane

The approximation of force-freeness pushes the magnetosonic point to infinity. This implies that there is a horizon beyond which the ZAMOs in the force-free region cannot observe the load region. Then, just as it is reasonable to presume the presence of the H-membrane at the stretched horizon, so it is also reasonable to presume the presence of the 'L-membrane' on the 'sphere at infinity' (Phinney 1982). The L-membrane can be endowed with similar physical properties, some of which turn out to be anti-symmetric, to those at the H-membrane. It is then possible to complete the discussion in the framework of force-freeness, avoiding the task of modelling the load region. As the H-membrane passes the 'energy-at-infinity' and angular momentum from the black hole to the force-free region with the rates expressed in (8.12) and (8.16), so also does the L-membrane from the force-free region to the load region.

The hypothetical L-membrane has surface charge density per unit area given from (4.1d) and (4.6);

$$\sigma_{\rm eL}=-(E_r/4\pi)=-(\Omega_{\rm F}/8\pi^2c)[\Psi_{\rm L}^{(2)}(\theta)/r^2]. \tag{8.20}$$

This charge density terminates all the electric flux that intersects the L-membrane. The total surface charge on the L-membrane becomes

$$Q_{\rm L}=\oint\sigma_{\rm eL}dA_{\rm L}=-(1/4\pi c)\int\Omega_{\rm F}(\theta)\Psi_{\rm L}^{(2)}(\theta)\sin\theta\,d\theta. \tag{8.21}$$

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We confine ourselves to uncharged Kerr black holes in this paper, so we take $Q_H = Q_L = 0$ in (8.5) and (8.21). There is a poleward surface electric current present on the L-membrane which is defined as a charge crossing a unit length perpendicular to $\mathcal{J}_{eL}$ per unit time [*cf.* (8.6)];

$$\mathcal{J}_{eL} = -[I(\Psi)/2\pi\varpi_L]\boldsymbol{e}_{\hat{\theta}} = (c/4\pi)\boldsymbol{E}_L, \qquad \boldsymbol{E}_L \equiv -[(\Omega_F\varpi/c)B_r]_L\,\boldsymbol{e}_{\hat{\theta}}, \tag{8.22a,b}$$

where (4.7a) and (4.12b) are used. This surface electric current flows across the field line labelled by Ψ against the surface electrical resistivity

$$R_L = 4\pi/c, \tag{8.23}$$

on the L-membrane according to Ohm's law. It is interesting to note that the electrical resistivities at both ends of the force-free field lines are equal to an impedance of free space at the end of an open wave guide (Thorne *et al.* 1986). The surface current also completes the circuit for all electric currents that intersect the L-membrane, and satisfies the charge conservation law

$$\partial\sigma_{eL}/\partial t + {}^{(2)}\nabla\cdot\mathcal{J}_{eL} + (-\boldsymbol{n}\cdot\boldsymbol{j}) = 0, \tag{8.24}$$

and Ampère's law

$$\mathcal{J}_{eL} = -(c/4\pi)(\boldsymbol{n}\times\boldsymbol{B}_L), \qquad \boldsymbol{B}_L = -|\boldsymbol{E}_L|\,\boldsymbol{e}_{\hat{\phi}} \tag{8.25a,b}$$

on the L-membrane, where the magnitude of $\boldsymbol{B}_L$ is given by (4.12b). Needless to say, expression (8.25a) is consistent with (8.22a) in view of the outgoing-wave boundary condition (4.13).

The L-membrane fields are degenerate in order of r^{-1}, but not force-free. The surface Lorentz force acting on the L-membrane is given by

$$\sigma_{eL}\boldsymbol{E}_L + (\mathcal{J}_{eL}/c)\times\boldsymbol{B}_n = [I(\Psi)/2\pi c\varpi]B_r\boldsymbol{e}_{\hat{\phi}} + (\Omega_F^2/c^2)[\Psi_L^{(2)}/8\pi^2 r^2](B_r\varpi^2)\boldsymbol{e}_{\hat{\theta}}. \tag{8.26}$$

The torque exerted by the electromagnetic field on the L-membrane per unit area per unit global time becomes, from (8.26),

$$dJ_L/dA_L\,dt = [I(\Psi)/2\pi c]B_r, \tag{8.27}$$

where J_L is the total angular momentum of the load region beyond the L-membrane. The load region therefore receives angular momentum from the force-free region with a rate

$$dJ_L/dt = (1/2\pi c)\oint I(\Psi)\,d\Psi, \tag{8.28}$$

where $d\Psi = B_r dA_L$ as before. This rate, of course, coincides with the rate in (8.12) at which the hole has lost its angular momentum.

One may presume that the ZAMOs living near the L-membrane are almost at rest without feeling the hole's gravity, i.e. $\Omega_L \simeq g_L \simeq 0$. One may then regard the surface temperature as negligibly small, but the surface area A_L is infinitely large and hence it is meaningful to consider the product of the surface gravity g_L and variation of the surface area, that is, the product of the membrane temperature and variation of the membrane entropy. If one denotes the total energy of the load region as $M_L c^2$, the first law of membrane thermodynamics and the energy–momentum relation for the L-membrane then yield

$$c^2 dM_L = T_L\,dS_L = \Omega_F dJ_L. \tag{8.29}$$

Then, for the change of the total energy per unit global time one obtains

$$c^2 dM_L/dt = P_L = (c/4\pi)\oint(\boldsymbol{E}\times\boldsymbol{B})_L\cdot\boldsymbol{n}\,dA_L = \oint R_L|\mathcal{J}_{eL}|^2\,dA_L = (1/2\pi c)\oint\Omega_F(\Psi)I(\Psi)\,d\Psi = P_H, \tag{8.30}$$

where P_L is the power from the force-free region to the load region. This is equal to the power P_H supplied by the hole through the H-membrane, that is, $dM_L/dt = -dM_H/dt$ must hold. The power extracted from the hole is invariant throughout the force-free region, because there is no dissipation of energy. Thus, the condition of $\zeta = 1$ also optimizes the power P_L from the force-free region to the load region. From (6.14a,b), (8.22a,b) and (8.25b), one has

$$|\boldsymbol{E}_L| \simeq |\boldsymbol{B}_L| \simeq (4\pi/c)|\mathcal{J}_{eL}| \simeq \zeta\Omega_H[(B_P\varpi^2)_H/c\varpi_L], \tag{8.31a}$$

for the case of $\zeta \ll 1$, and

$$|\boldsymbol{E}_L| \simeq |\boldsymbol{B}_L| \simeq (4\pi/c)|\mathcal{J}_{eL}| \simeq \Omega_H[(B_P\varpi^2)_H/c\varpi_L], \tag{8.31b}$$

for the case of $\zeta \gg 1$. It can be seen in (8.19a,b) and (8.31a,b) that both the membrane fields on the stretched and infinity horizons show similar dependence upon ζ.

In the force-free region the electromagnetic field completely governs the plasma motion and forces the $e^{\pm}$-particles, which have a velocity near the speed of light, to follow the field lines. Instead of explicitly taking into account finite inertia, we have presumed the presence of the L-membrane with electrical resistivity and poleward surface return current on it. The surface Lorentz force on the L-membrane possesses the non-vanishing poleward component. In reality, within the framework of force-freeness there is a plot of treason against electromagnetic fields, but for the sake of consistency we have suppressed inertial

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effects by making $\Psi_H^{(1)}(\lambda)=\Psi_L^{(1)}(\theta)=0$ in (4.6) and (5.5). It is obvious that one must somewhere include the effects from the outer light surface to infinity. Finite inertia will surely couple with the along-field acceleration of the $e^{\pm}$-particles, to lead to the transfield motion and the breakdown of charge separation. It will be of interest to see whether a general MHD wind will be collimated in the direction of the rotation axis by means of including plasma inertia and the pinching force of the toroidal field, to form astrophysical jets (*cf.*, Phinney 1982, 1983; Camenzind 1989; Heyvaerts & Norman 1989; Chineh, Li & Begelman 1991).

9 DC CIRCUIT THEORY

Znajek (1978b), Blandford (1979), Macdonald & Thorne (1982), Phinney (1982) and Thorne *et al.* (1986) showed that impedance matching, i.e. (load impedance) ≡ (source impedance), using circuit theory, produces a condition for optimal power outflow. However, these authors, except for Phinney (1982), did not use the critical condition (4.7b) or (6.6b) at the infinity horizon, but implicitly assumed the impedance properties on the L-membrane. Here, following Thorne *et al.*'s scheme, let us discuss the potential drops between the two horizons.

One can imagine a closed curve in the force-free magnetosphere that consists of *abcdef* (see Fig. 4); segment *ab* is on the stretched horizon with length Δl_H, segment *fc* is on the null surface where $\omega=\Omega_F$ and $\boldsymbol{E}+\boldsymbol{v}_F\times\boldsymbol{B}=0$, and segment *ed* with length Δl_L is on the infinity horizon. The lines *afe* and *bcd* are the magnetic surfaces and are labelled Ψ_1 and Ψ_2. One can effectively compute the voltage drops along the flux tube with $\Delta\Psi=\Psi_2-\Psi_1$. The total EMF around an arbitrary circuit is computed from Faraday's law:

$$\Delta V=\oint \alpha\boldsymbol{E}\cdot d\boldsymbol{l}=-(1/c)\oint(\boldsymbol{\beta}\times\boldsymbol{B})\cdot d\boldsymbol{l}=(1/2\pi c)\oint\omega\nabla\Psi\cdot d\boldsymbol{l}, \tag{9.1}$$

where $-\boldsymbol{\beta}\equiv\omega\varpi\boldsymbol{e}_{\hat{\phi}}$ is the ZAMOs rotational velocity (Thorne *et al.* 1986). Using the electric field given in (2.10c) measured by the ZAMOs, let us first consider the circuit $\subset_1$ consisting of curves *abcfa*, and then the EMF around the circuit $\subset_1$ becomes

$$\Delta V_1=\oint_{\subset_1}\alpha\boldsymbol{E}\cdot d\boldsymbol{l}=-(1/2\pi c)\oint(\Omega_F-\omega)\nabla\Psi\cdot d\boldsymbol{l}\simeq(1/2\pi c)(\Omega_H-\Omega_F)\Delta\Psi, \tag{9.2a}$$

because $\nabla\Psi\cdot d\boldsymbol{l}=0$ on segments *bc* and *fa* and $\boldsymbol{E}=0$ on segment *cf*. Next compute the EMF around the circuit $\subset_2$ consisting of *fcdef*;

$$\Delta V_2=\oint_{\subset_2}\alpha\boldsymbol{E}\cdot d\boldsymbol{l}=(1/2\pi c)\int_e^d\Omega_F d\Psi\simeq(1/2\pi c)\Omega_F\Delta\Psi, \tag{9.2b}$$

where $\omega=0$ on segment *ed* and (2.10a,c) is used. We have assumed that Δl_H and $\Delta\Psi$ are so small that Ω_F does not change so rapidly between Ψ_1 and Ψ_2 in the integration of (9.2a), and that, even though Δl_L is quite large or even diverges on the infinity horizon, the integration along segment *ed* converges to result (9.2b). Then, the total EMF around the circuit, $\subset\equiv\subset_1+\subset_2$, which is composed of curve *abcdef*, is given by

$$\Delta V=\Delta V_1+\Delta V_2\simeq(1/2\pi c)\Omega_H\Delta\Psi. \tag{9.3}$$

We can see in the last of expression (9.1) that, as long as the ZAMOs at infinity measure the voltage, that voltage on each surface with constant ω per unit flux tube is given by $(1/2\pi c)\omega$; the hole's horizon surface has $\Omega_H/2\pi c$, the null surface of $\omega=\Omega_F$ has $\Omega_F/2\pi c$, and the infinity horizon has zero voltage. The voltage drop, along the flux tube where $\Delta\Psi=\Psi_2-\Psi_1$ from the stretched horizon to the null surface, is given by ΔV_1, and the drop from the null surface to the infinity horizon is given by ΔV_2 and the total drop between the two horizons is given by (9.3). This voltage drop can be thought of as being maintained by a 'battery' in the hole's horizon.

The stellar wind theory, on the other hand, yields two critical conditions (6.6b) and (6.8b) for $I(\Psi)$, from which one has for the flux tube with $\Delta\Psi=\Psi_2-\Psi_1$,

$$I(\Psi)\simeq\begin{cases}\dfrac{(1/2\pi c)(\Omega_H-\Omega_F)\Delta\Psi}{(4\pi/c)(\Delta l_H/2\pi\varpi_H)}\equiv\dfrac{\Delta V_1}{\Delta R_H}, \\ \dfrac{(1/2\pi c)\Omega_F\Delta\Psi}{(4\pi/c)(\Delta l_L/2\pi\varpi_L)}\equiv\dfrac{\Delta V_2}{\Delta R_L},\end{cases} \tag{9.4a,b}$$

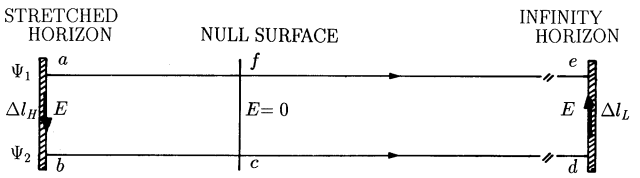


Figure 4. A schematic profile for an integration path $\subset=\subset_1+\subset_2$, which consists of the two field lines separated by $\Delta\Psi=\Psi_2-\Psi_1$ and the two segments with length Δl_H and Δl_L on the stretched horizon and the infinity horizon. The field lines are drawn as straight lines for simplicity.

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where $B_P=\Delta\Psi/2\pi\varpi\Delta l$, and $\Delta R_H=R_H\Delta l_H/2\pi\varpi_H$ and $\Delta R_L=R_L\Delta l_L/2\pi\varpi_L$ are the surface resistances between the two magnetic surfaces Ψ_1 and Ψ_2 in the stretched horizon and the infinity horizon, respectively. Then from (6.14a,b,c) and (9.4a,b), one has

$$\zeta=\Delta V_1/\Delta V_2=\Delta R_H/\Delta R_L. \tag{9.5}$$

The angular velocity of this flux tube is given by (6.14a). If one may take $\Delta l_L=(\varpi_L/\varpi_H)\Delta l_H$, i.e. $\zeta(\Psi)=1$ and $\Omega_F=\Omega_H/2$, one then has $\Delta R_H=\Delta R_L$ and hence $\Delta V_1=\Delta V_2$, which implies that the voltage drop from the H-membrane to the null surface is equal to that from the null surface to the L-membrane. Inversely, the condition for matching the two potential drops ΔV_1 and ΔV_2, which drive two centrifugal winds, means impedance matching on both the horizons, i.e. $\Delta R_H=\Delta R_L$, and hence $\Delta l_L=(\varpi_L/\varpi_H)\Delta l_H$ and $\zeta=1$. This corresponds to the optimal power extraction from the hole. The function ω decreases outwards from Ω_H on the H-membrane to zero on the L-membrane, naturally passing through the mid-surface of $\omega=(1/2)\Omega_H$. Then, the result of $\zeta=1$ indicates that the null surface where there must be the divider of neutral plasma must coincide with the mid-surface of $\omega=\Omega_H/2$.

One can derive other expressions for ΔV_1 and ΔV_2 from (9.4a,b),

$$\Delta V_1=(4\pi/c)[I(\Psi)/2\pi\varpi_H]\Delta l_H=R_H|\mathcal{J}_{eH}|\Delta l_H=|\boldsymbol{E}_H|\Delta l_H, \tag{9.6a,b}$$

$$\Delta V_2=(4\pi/c)[I(\Psi)/2\pi\varpi_L]\Delta l_L=R_L|\mathcal{J}_{eL}|\Delta l_L=|\boldsymbol{E}_L|\Delta l_L\ .$$

Thus, ΔV_1 and ΔV_2 can be interpreted as also meaning the voltage drops between the two magnetic surfaces Ψ_1 and Ψ_2 on the two horizon surfaces.

The electric current along the flux tube with $\Delta\Psi$ is given by $\Delta I=I(\Psi_2)-I(\Psi_1)=(dI/d\Psi)\Delta\Psi$. If the circuit under consideration lies in the region I (II), then the electric current flows from the L (H)-membrane to the H (L)-membrane without any dissipation on the way. If one considers the region I, electrons flow outward from the neutral domain and positrons flow inward from there. In the force-free model, the electric current is a centrifugal wind itself driven by the magnetic slingshot effect. The Joule heating arising in the stretched horizon and the infinity horizon gives rise to an increase of the hole's irreducible mass and particle acceleration respectively, in the load region. One can then see in (9.5) that $\Delta V_1\ll\Delta V_2$ and $\Delta R_H\ll\Delta R_L$ for $\zeta\ll1$, which imply that the hole's evolution path is given by $dS\simeq0$ from (8.15). For $\zeta\gg1$, one had, on the other hand, $\Delta V_1\gg\Delta V_2$ and $\Delta R_H\gg\Delta R_L$, and therefore the hole's evolution path can be approximated by $dM\simeq0$ from (8.16), as shown in detail in the next section (see also OK-I).

In neutron stars, surface ions are so strongly crystallized as to prevent being pulled out of the stars by the rotation-induced electric field; such that Ruderman & Sutherland (1975) suggested a pair-production mechanism for the magnetospheric plasma. Mestel, Phillips & Wang (1979) proposed a circulation model in which electrons are pulled out of the polar region of the neutron star's surface, migrate into the low-latitude zone after the along-field acceleration and dissipation due to γ-ray emission, and then return to the star along low-latitude field lines. In black holes, it is only by virtue of electrons (or positrons) falling into the hole that electric currents can flow out of (or into) the stretched horizon. The most economic way of maintaining the magnetospheric current system is by means of the charge-separated e^- or e^+ flow as stated in Section 6. Since field-aligned flow cannot reverse direction on field lines, there must be a fountain of $e^\pm$-plasma somewhere between the two horizons, most probably in the vicinity of the mid-surface at approximately the height of $0.6r_H$ above the true horizon. If the fountain dries up occasionally and the magnetic field lines threading the horizon are devoid of electric charges, there would be induced a quadrupole-like electric field around the mid-surface, which would be followed by discharges due to spark gaps and photon–photon collisions (Blandford & Znajek 1977). Otherwise the particles would be supplied by the pair creation due to collisions of photons from the accretion disc (Phinney 1982).

10 EVOLUTION OF KERR BLACK HOLES DUE TO EXTRACTION OF ROTATIONAL ENERGY

Equations (8.12), (8.15) and (8.16) are the conditions prescribing the thermodynamical and evolutional state of a Kerr black hole suffering from extraction of rotational energy and angular momentum through the force-free magnetosphere. Now, neglecting the dependence of ζ upon Ψ for brevity, but still regarding ζ as variable, let us work out the evolutional effects of ζ.

Equations (8.12), (8.15) and (8.16) combine to yield

$$-\Omega_H\frac{dJ_H}{dt}=\frac{1+\zeta}{\zeta}\left(T_H\frac{dS_H}{dt}\right)=-(1+\zeta)c^2\frac{dM_H}{dt}=\frac{\zeta}{1+\zeta}\frac{\Omega_H^2}{4\pi c}\oint(B_P\varpi^2)_H\,d\Psi. \tag{10.1}$$

It can be seen in (10.1) that a fraction $\zeta/(1+\zeta)=1-\Omega_F/\Omega_H$ of the loss rate of rotational energy is wasted in the Joule dissipation on the H-membrane, resulting in entropy increase, while a fraction $1/(1+\zeta)=\Omega_F/\Omega_H$ is given the output power. Also, one has

$$-\Omega_H\frac{dJ_H}{dt}\simeq T_H\frac{dS_H}{dt}\simeq\frac{\Omega_H^2}{4\pi c}\oint(B_P\varpi^2)_H d\Psi,\qquad c^2\frac{dM_H}{dt}\simeq-\frac{1}{\zeta}\frac{\Omega_H^2}{4\pi c}\oint(B_P\varpi^2)_H d\Psi \tag{10.2a,b}$$

for $\zeta\to\infty$, and

$$-\Omega_H\frac{dJ_H}{dt}\simeq-\frac{dM_H}{dt}c^2\simeq\zeta\frac{\Omega_H^2}{4\pi c}\oint(B_P\varpi^2)_H\,d\Psi,\qquad T_H\frac{dS_H}{dt}\simeq\zeta^2\frac{\Omega_H^2}{4\pi c}\oint(B_P\varpi^2)_H\,d\Psi \tag{10.3a,b}$$

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for $\zeta \to 0$. If one may make a simplifying assumption that $\oint (B_P \varpi^2)_H \, d\Psi$ remains constant for ζ, the former limit of $\zeta \to \infty$, i.e. $\Omega_F \to 0$, corresponds to the M_H-conservative process where almost all of rotational energy is wasted in the Joule dissipation with entropy production on the stretched horizon. The latter limit of $\zeta \to 0$, i.e. $\Omega_F \to \Omega_H$, on the other hand, corresponds to the S_H-conservative process where the output power is of smallness order ζ but the rate of entropy production is much smaller, of order ζ^2. The case of $\zeta = 1$ yields the process with equipartition between output power and surface entropy production (see also OK-I).

Let us now turn to clarifying a hole's evolution due to extraction of rotational energy. What is called the 'no-hair theorem' indicates that the physical state of any Kerr hole can be specified by only two independent variables, e.g. its mass and angular momentum. If as two such variables one chooses M_H and the dimensionless parameter h used in Section 7, one can express all other physical quantities in terms of M_H and h as given in (3.2a,b,...,l) in OK-I. We reproduce those which are necessary in the following:

$$S_H = (4\pi k_B G M_H^2/\hbar c)/(1+h^2), \qquad T_H = (\hbar c^3/8\pi k_B G M_H)(1-h^2), \qquad J_H = (2GM_H^2/c)[h/(1+h^2)], \tag{10.4a,b,...,f}$$

$$\Omega_H = (c^3/2GM_H)\,h, \qquad M_{irr} = M/(1+h^2)^{1/2}, \qquad M_{rot} = M[1-1/(1+h^2)^{1/2}].$$

It can be easily seen that $h = 0$ stands for the 'dead' Schwarzschild state and $h = 1$ stands for the 'most live' extreme Kerr state. Calculating $T_H(dS_H/dt)$ and $\Omega_H(dJ_H/dt)$, and combining the results with (8.12), (8.15) and (8.16), with use of (6.14a,b,c), one obtains

$$c^2 \frac{dM_H}{dt} = -\frac{\Omega_H^2}{4\pi c}\oint \frac{\zeta}{(1+\zeta)^2}(B_P \varpi^2)_H \, d\Psi = -P_H, \qquad \frac{(1-h^2)}{2(1+h^2)}\frac{dh^2}{dt} = -\frac{1}{M_H c^2}\frac{\Omega_H^2}{4\pi c}\oint \frac{\zeta}{(1+\zeta)^2}[(\zeta+1)+h^2(\zeta-1)](B_P \varpi^2)_H \, d\Psi, \tag{10.5a,b}$$

which are the basic equations describing the thermorotational evolution of a Kerr hole. Neglecting the dependence of ζ upon Ψ, one can then accomplish a kind of variable separation in (10.5a,b), that is,

$$\frac{1}{M_H}\frac{dM_H}{dh^2} = \frac{(1-h^2)}{2(1+h^2)[(\zeta+1)+(\zeta-1)h^2]},$$

$$\frac{(1-h^2)}{2(1+h^2)[(\zeta+1)+(\zeta-1)h^2]}\frac{dh^2}{dt} = -\frac{P_H}{M_H c^2}, \qquad P_H = \frac{\zeta}{(1+\zeta)^2}\frac{\Omega_H^2}{4\pi c}\oint (B_P \varpi)_H^2 \, dA_H. \tag{10.6a,b,c}$$

Taking $M_H = M_0$ for $h = h_0$ as an initial condition, one can easily integrate (10.6a) to obtain

$$M_H = M_0\left[\frac{1+h^2}{1+h_0^2}\right]^{1/2} F_M(h, h_0, \zeta), \qquad F_M(h, h_0, \zeta) = \begin{cases} \{[(\zeta+1)+(\zeta-1)h_0^2]/[(\zeta+1)+(\zeta-1)h^2]\}^{\zeta/2(\zeta-1)}, & \zeta \neq 1, \\ \exp[(h_0^2-h^2)/4], & \zeta = 1, \end{cases} \tag{10.7a,b}$$

where the case of $\zeta = 1$ is already given in (7.6) in OK-I. If a hole can evolve from quite near the extreme Kerr state with $h_0 \simeq 1$ to a Schwarzschild state with $h = 0$, then one has in the final state, from (10.7a,b),

$$M_f(\zeta) = M_0 \begin{cases} (1/2^{1/2})[2\zeta/(\zeta+1)]^{\zeta/2(\zeta-1)}, & \zeta \neq 1, \\ (1/2^{1/2})e^{1/4}, & \zeta = 1. \end{cases} \tag{10.8a,b}$$

For the extreme Kerr holes one has $M_{irr} = (1/2^{1/2})M_0$ and $M_{rot} = (1-1/2^{1/2})M_0$ from (10.4e,f) with $h = 1$. The behaviour of $M_f(\zeta)$ is depicted in Fig. 5, where one can see that $M_f(\zeta)/M_0$ ranges from unity for $\zeta \to \infty$, through 0.9673 for $\zeta = 4$ and 0.9079 for $\zeta = 1$, to 0.7071 for $\zeta = 0$. Thus, for $\zeta \to \infty$, one can confirm that the total mass energy is conserved [*cf.* (10.2b)], i.e. $M_f(\infty) = M_0$, with M_{rot} decreasing from near $(1-1/2^{1/2})M_0$ to zero and M_{irr} increasing from near $(1/2^{1/2})M_0$ to M_0. For $\zeta = 1$, while M_{rot} decreases to zero, M_{irr} increases from $(1/2^{1/2})\,M_0$ to $(e^{1/4}2^{1/2})\,M_0$, and $(1-e^{1/4}/2^{1/2})\,M_0$ is convected out of the horizon. For $\zeta \to 0$, the process becomes adiabatic, that is, $M_f(0) = M_{irr} = (1/2^{1/2})M_0$ and all of the hole's rotational energy is transferred through the magnetosphere to the load region with infinite time. For $\zeta = 4$, the total energy transported out of the horizon corresponds to 35 per cent of the optimal case of $\zeta = 1$, although the output power is 64 per cent of that in the case of $\zeta = 1$ (Phinney 1982). Then, as shown in Fig. 6, the evolutional paths on the J_H–Ω_H and S_H–T_H planes must then be situated inside the regions restricted by $M_H = M_0$ and $S_H = 2\pi G M_0^2/\hbar c$ (see also fig. 5 in OK-I).

One has already solved differential equation (10.6a) and obtained $M_H(h)$ as given in (10.7a,b), and hence one is aware of the dependence of other quantities upon ζ as well as h through (10.4a,b,...,f). If one nevertheless drops the ζ-dependence of Ω_H, it can easily be seen in (10.6c) that the output power is optimized by $\zeta = 1$, i.e. $\Omega_F = \Omega_H/2$, as shown by Blandford & Znajek (1977), Macdonald & Thorne (1982) and Thorne *et al.* (1986). Similarly, dropping the ζ-dependence of M_H and Ω_H, the time-scale of extraction will roughly be given from (10.6b,c);

$$t_{ext} \propto \frac{(1+\zeta)^2}{\zeta[(1+h^2)\zeta+1-h^2]}, \tag{10.9}$$

and $t_{ext} \propto \zeta^{-1} \to \infty$ for $\zeta \to 0$ and $dt_{ext}/d\zeta < 0$ for $0 < \zeta \leq \infty$. This implies that the extraction time becomes infinitely long in the

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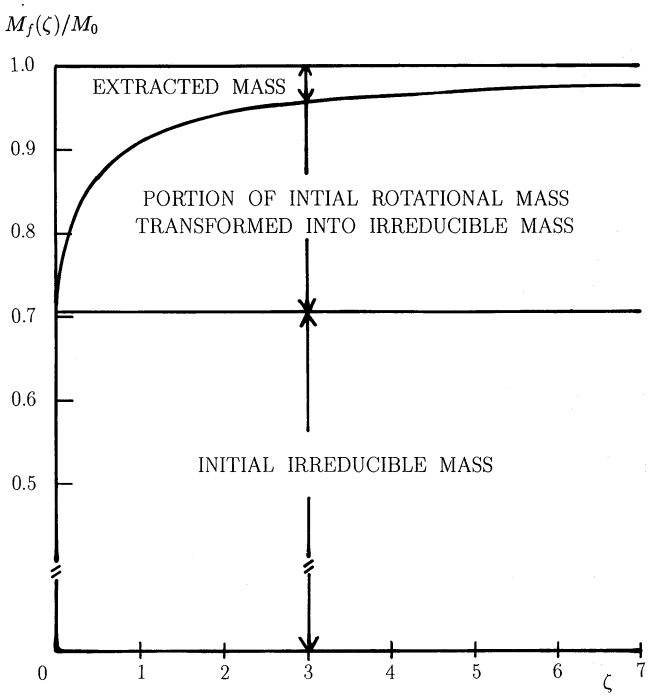


Figure 5. The dependence of the mass $M_f(\zeta)$ in the final Schwarzschild state upon parameter ζ. Note that $M_f(0)/M_0 = 1/2^{1/2}$ and $M_f(\infty)/M_0 = 1$.

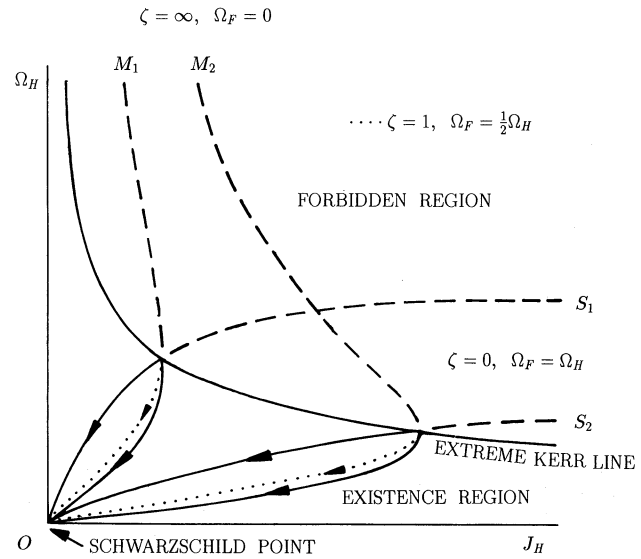


Figure 6. Evolutional paths of two holes with masses M_1 and M_2 in the angular-momentum–angular-velocity plane. The region outside the extreme Kerr line is the forbidden region. Under the condition of no supply of (irreducible) mass and angular momentum, the evolutional paths must lie in the region bounded by the two lines of S_H = constant ($\zeta \to 0$) and M_H = constant ($\zeta \to \infty$). The dotted lines indicate the paths for the equipartition process ($\zeta = 1$).

adiabatic limit of $\zeta \to 0$, in which all of a hole's rotational energy seems to be extractable, while t_{ext} tends to a finite, shortest time in the M_H-conservative limit of $\zeta \to \infty$.

In order to solve (10.6b,c) and to express evolutional parameter h in terms of global time t, one has to evaluate the variation of the strength of magnetic field on the H-membrane. Approximating B_P^2 with its average on the H-membrane for brevity, one has;

$$\oint (B_P \varpi)_H^2 \, dA_H \simeq \langle B_P^2 \rangle_H \oint \varpi_H^2 \, dA_H = \langle B_P^2 \rangle_H A_H \, r_H^2 f_A(h), \qquad f_A(h) \equiv \frac{1}{A_H \, r_H^2} \oint \varpi_H^2 \, dA_H = \left(\frac{1+h^2}{h}\right)^2 \left[\frac{1+h^2}{h} \tan^{-1} h - 1\right],$$

$$A_H \, r_H^2 = 4\pi r_H^4 (1+h^2) = A_0 r_0^2 \left(\frac{M_H}{M_0}\right)^4 \left(\frac{1+h_0^2}{1+h^2}\right)^3, \qquad A_0 r_0^2 \equiv 4\pi \left(\frac{2GM_0}{c^2}\right)^4 \frac{1}{(1+h_0^2)^3}, \qquad (10.10a,b,c)$$

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where $\varpi_H^2 = r_H^2(1+h^2)^2 \sin^2\theta/(1+h^2\cos^2\theta)$ and $dA_H = 2\pi r_H^2(1+h^2)\sin\theta\, d\theta$ are used. Suffix 0 denotes the initial value of each quantity at $t=0$. Function $f_A(h)$ changes by a factor of 3.4 from $f_A(1)=2\pi-4$ to $f_A(0)=2/3$. Then, from (10.6c), (10.7a,b) and (10.10a,b,c), one has for P_H;

$$P_H = P_0 \frac{\zeta}{(1+\zeta)^2} \frac{\langle B_P^2\rangle_H}{\langle B_P^2\rangle_0} \frac{[(1+h^2)/h]\tan^{-1}h - 1}{[(1+h_0^2)/h_0]\tan^{-1}h_0 - 1} F_M(h, h_0, \zeta)^2, \qquad P_0 \equiv \frac{\Omega_{H,0}^2}{4\pi c}\langle B_P^2\rangle_0 A_0 r_0^2 f_A(h_0), \tag{10.11a,b}$$

where $\langle B_P^2\rangle_0$ stands for the initial value or an appropriate normalization factor for $\langle B_P^2\rangle_H$. Combining (10.11a,b) with (10.6b), one has

$$H_0 \int_h^{h_0} \frac{(1+h^2)^{3/2}(1-h^2)}{h f_A(h) F_M(h, h_0, \zeta)[(\zeta+1)+(\zeta-1)h^2]}\, dh = \frac{1}{t_0}\frac{\zeta}{(1+\zeta)^2}\int_0^t \frac{\langle B_P^0\rangle_H}{\langle B_P^2\rangle_0}\, dt, \qquad t_0 \equiv \frac{M_0 c^2}{P_0}, \qquad H_0 \equiv \frac{h_0^2 f_A(h_0)}{(1+h_0^2)^{5/2}}. \tag{10.12a,b,c}$$

Then, from (10.12a), (10.7b) and (10.10b), one has

$$\frac{1}{t_0}\int_0^t \frac{\langle B_P^2\rangle_H}{\langle B_P^2\rangle_0}\, dt = H_0 \int_h^{h_0} \frac{h\, dh}{[(1+h^2)/h]\tan^{-1}h - 1} \times \begin{cases} \dfrac{1}{b_0^{1/2}}\left(\dfrac{1-h^2}{1+h^2}\right)^{1/2}, & \text{for } \zeta\to\infty, \\ \dfrac{2}{e^{h_0^2/4}}\dfrac{(1-h^2)e^{h^2/4}}{(1+h^2)^{1/2}}, & \text{for } \zeta = 1, \\ \dfrac{1}{\zeta}\dfrac{1}{1+h^2}, & \text{for } \zeta\to 0. \end{cases} \tag{10.13a,b,c}$$

where $b_0 \equiv 1 - h_0^2$ for $h_0 \neq 1$ and $\equiv 2$ for $h_0 = 1$ from (10.7b).

It can be seen in (10.12a) and (10.13a,b,c) that except for the limiting case of $\zeta\to 0$, the integrand of the right-hand side of (10.12a) contains a factor of $(1-h^2)$ for $0<\zeta<\infty$ [see also (10.6b)]. Thus if $\langle B_P^2\rangle$ is finite in the extreme Kerr state, although this is not far from obvious [see e.g. Bičák & Janiš (1985)], one has $(1-h^2)\, dh/dt \simeq$ constant in the vicinity of $h=1$ for $0<\zeta<\infty$. This implies that dh/dt has infinite negative gradient at $h\simeq 1$. One can also see in (10.12a) that h will have an exponential tail for $t\to\infty$, if $\langle B_P^2\rangle$ remains finite for $h\to 0$.

If one presumes the situation that the magnetic pressure of the hole's magnetosphere is kept nearly constant, in balance with that of the accretion disc, i.e. $\langle B_P^2\rangle_H \simeq \langle B_P^2\rangle_0$, the results of numerical integration are shown in Fig. 7(a)–(b), where $h(t)$ and the t-dependence of other physical quantities for $h_0=1$ and $\zeta=1$ are portrayed. If one may make such a rough approximation as $\langle B_P^2\rangle_H \simeq \langle B_P^2\rangle_0(1-h^2)$ for the behaviour of $\langle B_P^2\rangle_H$ near the extreme Kerr state, $h(t)$ will have finite slope at the initial phase of $h\simeq 1$ or $t=0$. Then a vanishingly small flux of magnetic field may be enough for the initial phase of braking a near-extreme Kerr hole. Details need further extended numerical simulations.

One can now make some numerical estimations for a near-extreme Kerr black hole with $M_0 \sim 10^9\, M_\odot$ and $\langle B_P^2\rangle_0^{1/2} \sim 10^4$ G. For the initial values of the hole's physical quantities and for $h_0 \simeq 1$, one has;

$$r_{H,0} = a_0 = 1.48\times 10^{14}(M_0/10^9 M_\odot)\ \text{cm}, \quad g_{H,0} = 1.52\times 10^6 (M_0/10^9 M_\odot)^{-1}\ \text{cm s}^{-2}, \quad T_{H,0} = 6.16\times 10^{-17}(M_0/10^9 M_\odot)^{-1}\ \text{K},$$
$$A_{H,0} = 5.49\times 10^{29}(M_0/10^9 M_\odot)^2\ \text{cm}^2, \quad S_{H,0} = 7.26\times 10^{78}(M_0/10^9 M_\odot)^2\ \text{erg deg}^{-1}, \quad \Omega_{H,0} = 1.01\times 10^{-4}(M_0/10^9 M_\odot)^{-1}\ \text{s}^{-1},$$
$$J_{H,0} = 8.82\times 10^{66}(M_0/10^9 M_\odot)^2\ \text{erg s}^{-1}. \tag{10.14a,b,...,g}$$

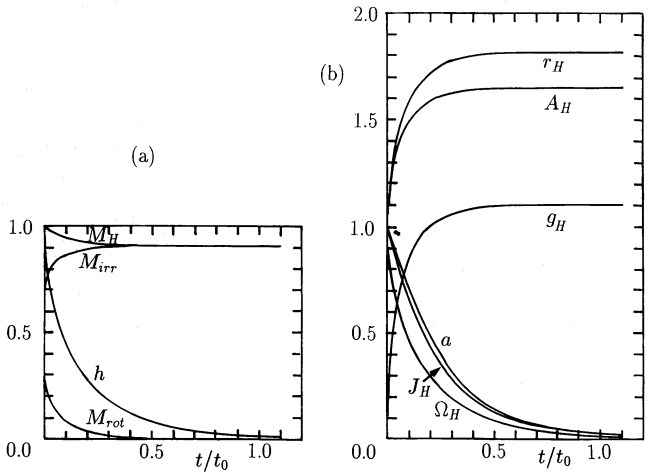


Figure 7. (a) Changes of the evolutional parameter h and the hole's three masses M_H, M_{irr} and M_{rot} through h with the global time t. (b) Changes of other physical quantities r_H, A_H, g_H, a, J_H and Ω_H through h with the global time t.

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Precisely speaking, a factor $(1-h_0^2)\simeq 0$ should be attached to the initial values of g_H and T_H. One can easily check $(\Omega_H J_H)_0 = 1/2 M_0 c^2 = 8.95\times 10^{62}(M_0/10^9 M_\odot)$ erg.

Making use of the above estimates, one obtains, from (10.10a) and (10.11a);

$$P_0 = 2.34\times 10^{45}[\langle B_P^2\rangle_0/(10^4\ \mathrm{G})^2](M_0/10^9 M_\odot)^2\ \mathrm{erg\ s^{-1}}, \qquad t_0 = 2.42\times 10^{10}[\langle B_P^2\rangle_0/(10^4\ \mathrm{G})^2]^{-1}(M_0/10^9 M_\odot)^{-1}\ \mathrm{yr}. \tag{10.15a,b}$$

For the surface electric current flowing equatorwards on the H-membrane, one can obtain in the order of magnitude,

$$\langle|\mathcal{J}_H|^2\rangle^{1/2} \simeq (P_H/R_H A_H)^{1/2} \simeq 1.50\times 10^3\ \mathrm{A\ cm^{-1}} \tag{10.16}$$

for $h_0\simeq 1$, where we used $R_H = 377$ Ohm. The electromotive force maintaining the power is given by

$$EMF = (\Omega_H/2\pi c)\oint d\Psi \simeq (\Omega_H/2\pi c)\langle B_P\rangle A_H = 2a\langle B_P\rangle, \tag{10.17a}$$

and at the initial state with $h_0\simeq 1$, one has

$$EMF_0 = 8.86\times 10^{20}(\langle B_P\rangle_0/10^4\ \mathrm{G})(M_0/10^9 M_\odot)\ \mathrm{V}. \tag{10.17b}$$

It thus turns out that one can pursue almost analytically the hole's evolution from the most active phase to the dying condition.

11 DISCUSSION AND CONCLUSION

Before ending this paper, it will be worthwhile mentioning discrepancies at two significant points between the physics in this article and the recent research due to Punsley & Coroniti (1990) (see Punsley 1991, references therein). Punsley (1991) claims that an accreting, magnetically dominated plasma flow will in general be dissipative, and claims that early work on this topic in Blandford & Znajek (1977), Macdonald & Thorne (1982) and Phinney (1982, 1983) did not think of the plasma inflow as dissipative. The membrane paradigm was introduced by Thorne *et al.* (1986) as a viewpoint which was naturally developed on the same line as these authors with respect to black hole electrodynamics. In a 'frozen-star-like' view so far of physics outside the horizon, particles and fields very near the horizon possess a highly complex, frozen, 'boundary layer' structure, which is essentially a relic of the black hole's past. The membrane viewpoint 'stretches' the horizon to cover up the boundary layer and imposes membrane-like boundary conditions on the stretched horizon. This membrane is endowed with electrical resistivity as well as viscosity, together with other physical properties. As shown in previous sections, the surface electric current is Joule-dissipated in the stretched horizon with electrical resistivity, resulting in the entropy increase of the hole, and the latter can also be interpreted as a consequence of the ingoing infinitely blueshifted Poynting flux. This is consistent with the first and second laws of black hole thermodynamics. The outgoing Poynting flux flows into the 'sphere-at-infinity', of which the electrical resistivity dissipates the surface return current. The present work is to further deepen the understanding of physics outside the horizon than these authors, and yet we can say that they contain implicitly or explicitly the dissipative nature of plasma flow in their analyses. In this idealized treatment of a force-free magnetosphere, based upon the membrane paradigm, we have only to take into account the dissipative process due to Joule heating on the H- and L-membrane.

The second point is the question of causality. Punsley (1991) insists that the event horizon behaves like a vacuum infinity to accreting, charge-neutral plasma flows, and the horizon is like $\varpi\to\infty$, except for the following two differences; one is that the angular velocity of the horizon is Ω_H and the other is that the horizon infinity is a compact set. There is a third, yet crucial difference which Punsley dismissed. This is that a Kerr black hole does communicate, beyond the 'horizon infinity', its mass to infinity through the g_{00} component of the metric and its angular velocity through the $g_{0\phi}$ component, and the plasma inflow is equivalent to a centrifugal wind induced by the effect of inertial-frame dragging, that is the $\alpha\omega$ mechanism. We have, however, now no more comment on this point than saying that as long as a black hole is magnetically dominated, the stretched horizon will still keep causal contact with the outgoing wind (see Blandford 1989).

In this paper, we studied the force-free black hole magnetosphere, with the aid of Macdonald & Thorne's (1982) electrodynamics based upon the '3+1' formalism and Thorne *et al.*'s (1986) paradigm view. The assumption of force-freeness and charge separation enables us to understand the fundamental properties of magnetospheric structure, the process of energy extraction via the centrifugal-wind–electric-current system and the hole's resulting evolution. The 'force-free' magnetosphere is just a kind of device with which to equip a Kerr hole, to pump out the hole's energy to infinity, and hence we did not discuss how the extracted energy in the form of electromagnetic power should be transferred to particle energy, i.e. particle acceleration and jet formation. However, even within the framework of force-freeness, we find that there is some relation between the power extraction rate, i.e. evolutional time-scale and parameter ζ expressing the collimation of field lines. As ζ increases, the collimation will be strengthened. Also, there is a non-vanishing surface Lorentz force acting polewards on the L-membrane between the force-free region and the load region. These facts suggest that particle acceleration and jet formation will have profound effects upon the hole's magnetosphere and evolution. In order to discuss the interactions of the black hole with its environment, it goes without saying that one needs a magnetohydrodynamic treatment and also inclusion of the accretion disc.

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APPENDIX A: $\nabla \cdot E$, CHARGE DENSITY AND TOROIDAL ELECTRIC CURRENT

In calculations very near the horizon the coordinate system (α, λ, ϕ) with its metric given in (2.5) is useful. One can express the Boyer–Lindquist quantities defined in (2.1) in order of α^2 as follows:

$$r \simeq r_H + (\Sigma_H^2/2\rho_H^2 r_H)\alpha^2, \qquad \rho^2 \simeq \rho_H^2 + (\Sigma_H^2/\rho_H^2)\alpha^2, \qquad \Delta \simeq (\Sigma_H^2/\rho_H^2)\alpha^2, \qquad \Sigma^2 \simeq [1 + (1 + \Sigma_H/\rho_H^2)\alpha^2],$$

$$\omega \simeq \Omega_H\{1 - [1 + \Sigma_H(r_H^2 - a^2)/2\rho_H^2 r_H^2]\alpha^2\}, \qquad \varpi \simeq [1 + (1/2)(1 - \Sigma_H a^2 \sin^2\theta/\rho_H^2)\alpha^2], \tag{A1}$$

where $\rho_H^2 = r_H^2 + a^2\cos^2\theta$, $\Sigma_H = r_H^2 + a^2$ and $\varpi_H = (\Sigma_H/\rho_H)\sin\theta$. Calculating the divergence of $\boldsymbol{E}$ with the use of (5.2b) and (A1),

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one has

$$\nabla\cdot\boldsymbol{E}\simeq\frac{g_{\rm H}}{c^2}\frac{1}{\varpi}\frac{\partial}{\partial\alpha}(\varpi E_r)+\frac{1}{\varpi}\frac{\partial}{\partial\lambda}(\varpi E_\theta)\simeq\frac{1}{2\pi c\alpha(\Omega_{\rm H}-\Omega_{\rm F})}\left\{\frac{g_{\rm H}^2}{c^4}(\Omega_{\rm H}-\Omega_{\rm F})^2\,\alpha\frac{\partial}{\partial\alpha}\left(\frac{1}{\alpha}\frac{\partial\Psi}{\partial\alpha}\right)+\frac{1}{2\varpi^2(\partial\Psi/\partial\lambda)}\frac{\partial}{\partial\lambda}\right.\tag{A2a}$$

$$\left.\times\left[\varpi(\Omega_{\rm H}-\Omega_{\rm F})\frac{\partial\Psi}{\partial\lambda}\right]+A\right\},$$

$$A\equiv\frac{g_{\rm H}^2}{c^4}\frac{1}{2\varpi^2}\frac{\partial}{\partial\alpha}[\varpi(\omega-\Omega_{\rm H})]^2\simeq\frac{g_{\rm H}^2}{c^4}\left\{-(\Omega_{\rm H}-\Omega_{\rm F})\frac{d\Omega_{\rm F}}{d\Psi}\left(\frac{\partial\Psi}{\partial\alpha}\right)^2+\left(\alpha\frac{\partial\Psi}{\partial\alpha}\right)\left[(\Omega_{\rm H}-\Omega_{\rm F})^2\left(1-\frac{\Sigma_{\rm H}a^2\sin^2\theta}{\rho_{\rm H}^4}\right)\right.\right.\tag{A2b}$$

$$\left.\left.-2\Omega_{\rm H}(\Omega_{\rm H}-\Omega_{\rm F})\left\{1+\frac{\Sigma_{\rm H}(r_{\rm H}^2-a^2)}{2\rho_{\rm H}^2r_{\rm H}^2}\right\}\right]+O(\alpha^2)\right\}.$$

One can also evaluate the values of $\varrho_{\rm e}$ and $j_{\rm T}$ near the horizon, from (3.2a) and (A1):

$$\alpha\varrho_{\rm e}\simeq\frac{(dI^2/d\Psi)}{c\varpi^2(\Omega_{\rm H}-\Omega_{\rm F})}+\frac{c\alpha^2}{\varpi^4(\Omega_{\rm H}-\Omega_{\rm F})^3}\left(\frac{dI^2}{d\Psi}\right)-\frac{c}{8\pi^2\varpi^2(\Omega_{\rm H}-\Omega_{\rm F})}B,\tag{A3a}$$

$$B\equiv\frac{2\alpha^2}{c^2}\nabla\Psi\cdot\boldsymbol{G}\simeq\frac{g_{\rm H}^2}{c^4}\left(2\alpha\frac{\partial\Psi}{\partial\alpha}\right)\left\{1-\frac{\varpi_{\rm H}^2}{c^2}\Omega_{\rm H}(\Omega_{\rm H}-\Omega_{\rm F})\left[1+\frac{\Sigma_{\rm H}(r_{\rm H}^2-a^2)}{2\rho_{\rm H}^2r_{\rm H}^2}\right]\right\}+O(\alpha^2).\tag{A3b}$$

Combining (A2a,b) with (2.11), or directly from (3.2b), one has

$$j_{\rm T}\simeq\frac{g_{\rm H}^2}{4\pi^2c^3\varpi}\left(\frac{1}{\alpha}\right)\frac{\partial\Psi}{\partial\alpha}\left\{1-\frac{\varpi_{\rm H}^2}{c^2}\Omega_{\rm H}(\Omega_{\rm H}-\Omega_{\rm F})\left[1+\frac{\Sigma_{\rm H}(r_{\rm H}^2-a^2)}{2\rho_{\rm H}^2r_{\rm H}^2}\right]\right\}+O(1).\tag{A4}$$

Thus one can obtain the expressions of $\nabla\cdot\boldsymbol{E}$ and $\varrho_{\rm e}$ to the order of α, and that of $j_{\rm T}$ to the order of α^{-1}.

APPENDIX B: EFFECTS OF MAGNETIC CHARGE AND CURRENT

In the text we have followed Damour–Thorne *et al.*'s version in formulating the four electric laws of the stretched horizon (Thorne *et al.* 1986), and those are the laws of Gauss (8.4), Ampère (8.7), Ohm (8.8) and charge conservation (8.9). Znajek's version of the horizon's electromagnetic laws (see Thorne *et al.* 1986) differs from Damour–Thorne *et al.*'s version adopted so far in that Znajek endows the horizon with magnetic charge as well as electric charge and very high volume conductivities for both magnetic and electric currents. It will be interesting to see that Damour–Thorne *et al.*'s formalism can be modified to include the surface magnetic charge and current and this has the beauty and advantage of treating $\boldsymbol{E}$ and $\boldsymbol{B}$ on a completely equal footing.

Maxwell's equations generalized to comprise the magnetic charge and current are invariant with respect to the interchange

$$\boldsymbol{E}\rightarrow\boldsymbol{B},\qquad\boldsymbol{B}\rightarrow-\boldsymbol{E},\qquad\varrho_{\rm e}\rightarrow\varrho_{\rm m},\qquad\varrho_{\rm m}\rightarrow-\varrho_{\rm e},\qquad\boldsymbol{j}_{\rm e}\rightarrow\boldsymbol{j}_{\rm m},\qquad\boldsymbol{j}_{\rm m}\rightarrow-\boldsymbol{j}_{\rm e},\tag{B1}$$

(see e.g. Schwinger 1969), where suffixes e and m denote 'electric' and 'magnetic' as before. One can then see that even in the presence of any real charges and currents there is a complete duality between electricity and magnetism. Similarly to the definition of $\sigma_{\rm eH}$, the surface magnetic charge per unit area is the one required to terminate the magnetic field lines which thread the stretched horizon;

$$\sigma_{\rm mH}\equiv(B_{\rm n}/4\pi)_{\rm H}=(1/\pi^2\varpi_{\rm H})[d\Psi_{\rm H}^{(0)}/d\lambda].\tag{B2}$$

The total surface magnetic charge on the H-membrane becomes zero, because

$$\oint\sigma_{\rm mH}\,dA_{\rm H}=(1/4\pi)\oint d\Psi=(1/4\pi)[\Psi(\text{north pole})-\Psi(\text{south pole})]=0,\tag{B3}$$

where (2.6a) and (B2) are used. The surface magnetic current is defined as terminating all tangential electric field lines

$$\boldsymbol{E}_{\rm H}=-(4\pi/c)\mathcal{J}_{\rm mH}\times\boldsymbol{n}.\tag{B4}$$

This condition combines with the 'radiative boundary condition' (8.2), to yield Ohm's law with the same resistivity as given in (8.8)

$$\mathcal{J}_{\rm mH}=(c/4\pi)\boldsymbol{B}_{\rm H}\tag{B5a}$$

for the relation between the toroidal magnetic field and current. This result indicates that the surface magnetic current on the H-membrane should be conduction current, but making use of the critical condition (5.6a) and (8.1b), (B2), (B5a) and (2.8), one

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obtains

$$\mathcal{J}_{mH} = -(B_n/4\pi)\,\varpi_H(\Omega_H - \Omega_F)\,\boldsymbol{e}_{\hat{\phi}} = \sigma_{mH}(\alpha \boldsymbol{v}_F)_H, \tag{B5b}$$

which implies that the current is also a convection current of magnetic charge σ_{mH} due to rotation in the negative direction of the field lines related as measured using global time by a ZAMO. The total current crossing a longitude from the north pole to the south pole on the H-membrane is given by

$$\int \mathcal{J}_{mH}\, d\lambda = -(1/8\pi^2)\int (\Omega_H - \Omega_F)\, d\Psi\, \boldsymbol{e}_{\hat{\phi}}, \tag{B6}$$

which may not vanish unless Ω_F is independent of Ψ [*cf.* (B3)]. If there is no inflow of magnetic charge, it is easy to see that σ_{mH} and $\mathcal{J}_{mH}$ satisfy a similar law of magnetic charge conservation as in (8.9);

$$\partial \sigma_{mH}/\partial t + {}^{(2)}\nabla \cdot \mathcal{J}_{mH} = 0. \tag{B7}$$

In view of the interchange law (B1), there is complete symmetry between the four laws about σ_{eH}, $\mathcal{J}_{eH}$ and the four laws about σ_{mH}, $\mathcal{J}_{mH}$.

If one presumes the presence of the magnetic charge and current on the stretched horizon, just as the electromagnetic field exerts the electric Lorentz force on the electric charge and current, so also should it exert the magnetic Lorentz force on the magnetic charge and current. The expression for the magnetic Lorentz force is obtainable from that for the electric Lorentz force (8.10) with the use of the interchange laws (B1). In the force-balance equation on the stretched horizon, one should then take into account the magnetic as well as the electric Lorentz force, that is,

$$[\sigma_{eH}\boldsymbol{E}_H + (\mathcal{J}_{eH}/c) \times \boldsymbol{B}_n] + [\sigma_{mH}\boldsymbol{B}_H - (\mathcal{J}_{nH}/c) \times \boldsymbol{E}_n] \tag{B8}$$

[see Schwinger (1969)]. Making use of (B2), (8.7), (B5a) and (8.2), one can then show that

$$\sigma_{eH}\boldsymbol{E}_H = -(\mathcal{J}_{mH}/c) \times \boldsymbol{E}_n, \qquad \sigma_{mH}\boldsymbol{B}_H = (\mathcal{J}_{eH}/c) \times \boldsymbol{B}_n. \tag{B9}$$

Thus, the force that the poloidal electric field produces on the surface electric charge is equivalent to the force produced as the surface toroidal magnetic current flows across the normal electric field. Also the force that the toroidal magnetic field produces on the surface magnetic charge is equivalent to the force produced as the surface poloidal electric current flows across the normal magnetic field. In short, the magnetic Lorentz force is exactly equal to the electric Lorentz force. The equivalence of both the Lorentz forces is brought about by the radiative boundary condition and magnetic as well as electric Ohm's laws on the H-membrane. Then, if one takes into account the presence of magnetic charge and current as well as electric charge and current, one has twice the electric Lorentz force for the total force on the stretched horizon, and hence double the extraction rates of rotational energy and angular momentum from black holes.

Maxwell's equation and the total surface Lorentz force (B8) are invariant for the duality transformation

$$\boldsymbol{E}'_H = \boldsymbol{E}_H \cos\chi + \boldsymbol{B}_H \sin\chi, \qquad \mathcal{J}'_{eH} = \mathcal{J}_{eH}\cos\chi + \mathcal{J}_{mH}\sin\chi, \qquad \sigma'_{eH} = \sigma_{eH}\cos\chi + \sigma_{mH}\sin\chi, \tag{B10a,b}$$

$$\boldsymbol{B}'_H = -\boldsymbol{E}_H \sin\chi + \boldsymbol{B}_H \cos\chi, \qquad \mathcal{J}'_{mH} = -\mathcal{J}_{eH}\sin\chi + \mathcal{J}_{mH}\cos\chi, \qquad \sigma'_{mH} = -\sigma_{eH}\sin\chi + \sigma_{mH}\cos\chi.$$

One can make a duality transformation, choosing the angle χ so that $\sigma'_{mH} = 0$ and $\mathcal{J}'_{mH} = 0$, and then have Maxwell's equations and the total surface Lorentz force be formally comprised of no magnetic charge and current (Schwinger 1969; Jackson 1975). That is to say, in a coordinate system with angle χ so chosen, one has $\boldsymbol{f}'_m = 0$ and

$$\boldsymbol{f} = \boldsymbol{f}'_e = \sigma'_{eH}\boldsymbol{E}'_H + (\mathcal{J}'_{eH}/c) \times \boldsymbol{B}'_n. \tag{B11}$$

However, this will not change the fact that the electric Lorentz force exerted on the stretched horizon is twice as large as the case where the magnetic Lorentz force is not at all taken into account.

Quite analogously, one can examine the effects of the hypothetical magnetic charge and current on the L-membrane as well. Similarly to σ_{mH} and $\mathcal{J}_{mH}$, these are defined

$$\sigma_{mL} = -(B_r/4\pi)_L, \qquad \mathcal{J}_{mL} = (c/4\pi)(\boldsymbol{n} \times \boldsymbol{E}_L) = (c/4\pi)\boldsymbol{B}_L = \sigma_{mL}(\boldsymbol{v}_F)_L, \tag{B12a,b}$$

where (4.13) and (2.8) are used and $\boldsymbol{n}$ is the unit outwards normal vector. One can see that the total magnetic charges vanish, i.e.

$$\oint \sigma_{mL}\, dA_L = -(1/4\pi)\oint d\Psi = 0 \tag{B13}$$

on the L-membrane as well as on the H-membrane [*cf.* (B3)]. Expression (B12b) indicates that the hypothetical magnetic current is not only conduction current with conductivity $c/4\pi$ but also convection current of the surface magnetic charge σ_{mL} due to rotation of magnetic field lines. The 'magnetic' Lorentz force acting on the L-membrane can be shown to be equal to the 'electric' one given in (8.26), i.e.

$$-(\mathcal{J}_{mL}/c) \times \boldsymbol{E}_n + \sigma_{mL}\boldsymbol{B}_L = \sigma_{eL}\boldsymbol{E}_L + (\mathcal{J}_{eL}/c) \times \boldsymbol{B}_n. \tag{B14}$$

Then, the total surface force due to both the electric and magnetic Lorentz forces is double that of the electric Lorentz force.

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1992MNRAS.254..1920

It is thus possible to endow the stretched horizon with magnetic charge as well as electric charge, and with the same surface conductivities for both magnetic current and electric current. The resulting charges and currents terminate all electromagnetic quantities in a thin skin just above the horizon. Inclusion of magnetic charge and current produces double the surface Lorentz force than is true without them, thereby enabling the transfer of the hole's energy and angular momentum to the force-free region at double the rate. Then, unless the force-free region can store energy and angular momentum, in the stationary state it must transfer the same quantities of energy and angular momentum gained at the stretched horizon through the force-free region to the infinity horizon. This will require one to include the effects due to the surface magnetic charge and current even on the L-membrane which will be situated in much flatter space-time where it is evident that magnetic monopoles are non-existent. Therefore, in spite of the quite attractive possibility that the inclusion of magnetic charge and current enables one to extract the hole's energy and angular momentum at double the rates, in the text we have adopted the formalism without postulating the existence of magnetic charge and current (*cf.* Thorne *et al.* 1986).

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